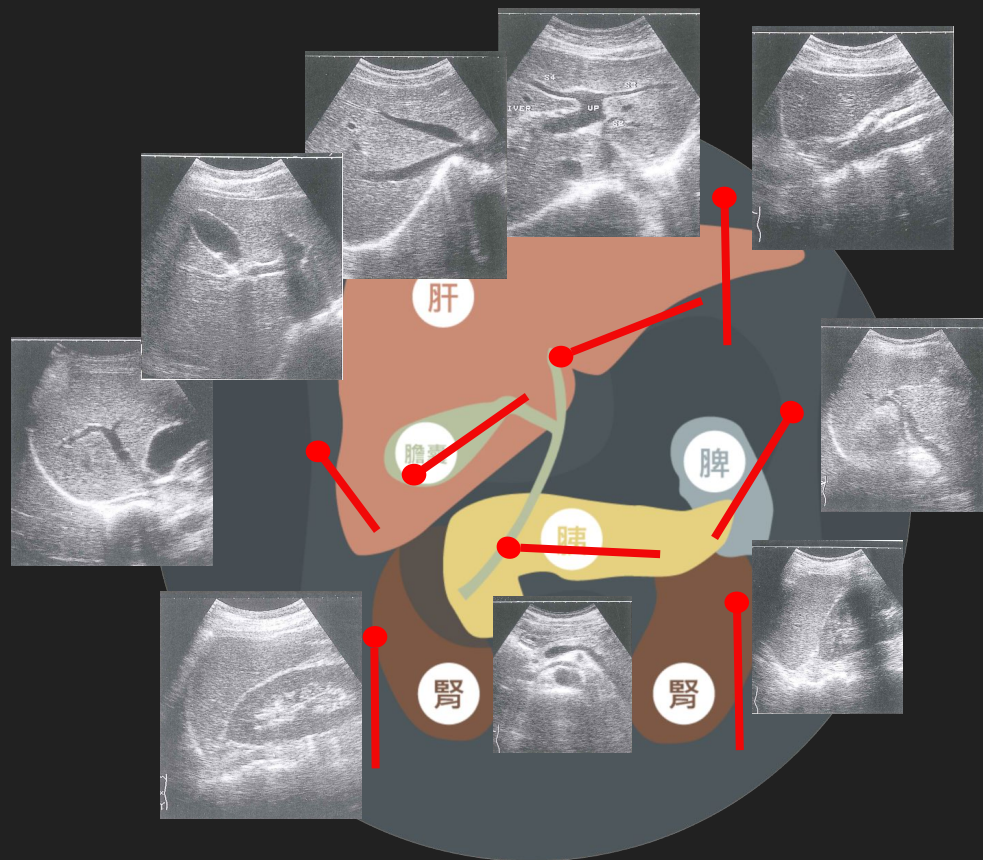


Problem-oriented approach to abdominal ultrasound

F1 郭加智

必掃清單

- Liver edge
- 工字型(左肝)
- Hepatic v
- Portal triad
- 膽囊
- 腎臟
- 脾臟
- 胰臟



第一篇章：正常的解剖構造

基本view

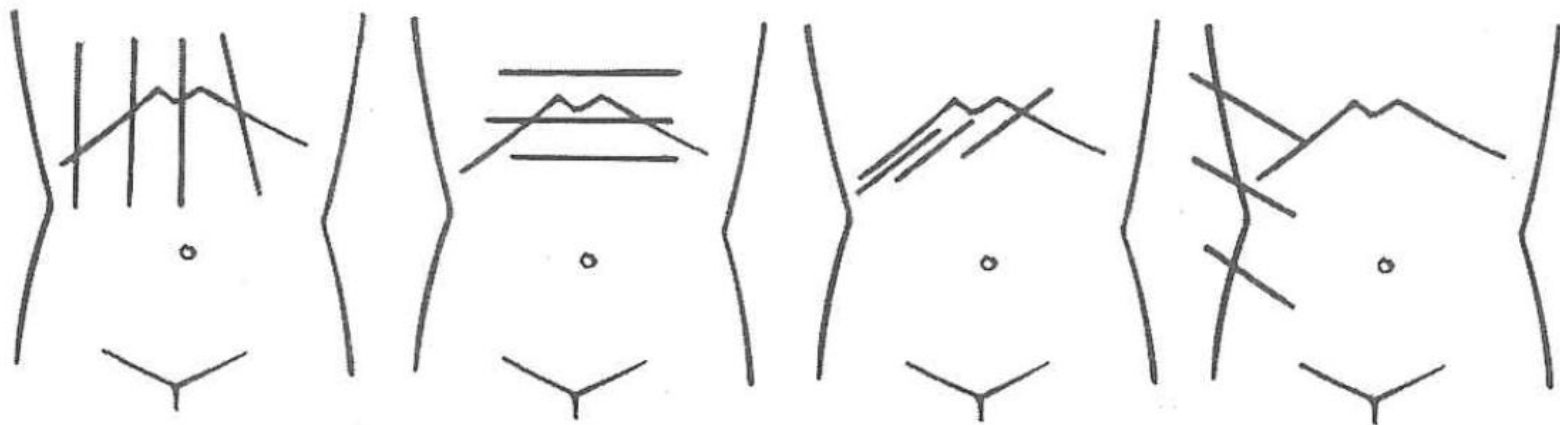
肝

膽

胰

腎脾

掃描重點 View



基本姿勢

背臥姿

背臥姿

背臥姿

背臥姿

掃描

縱向切面掃描
(sagittal scan)

橫向切面掃描
(transverse scan)

右肋骨弓下掃描
肝臟左葉至右葉，

右肋間掃描
肝右葉，膽囊，

目標器官

肝臟左葉至右葉

肝臟左葉

膽囊

右腎

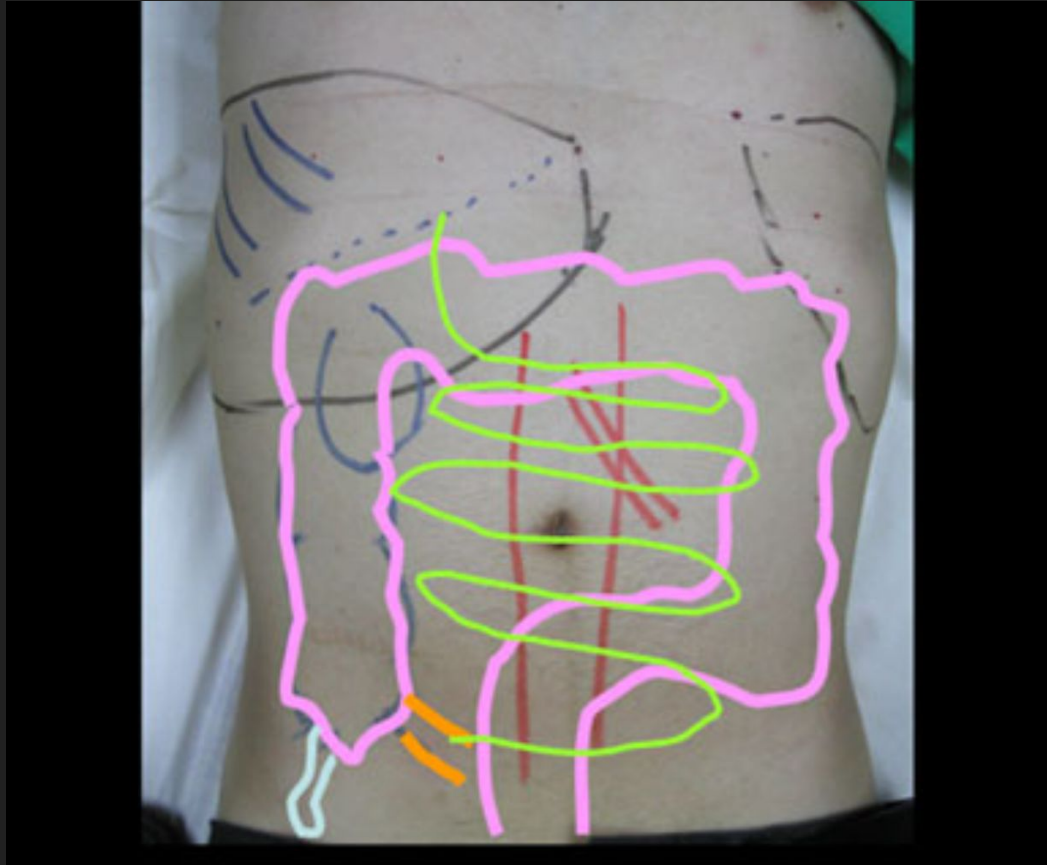
Sagital

Transverse

Subchondral

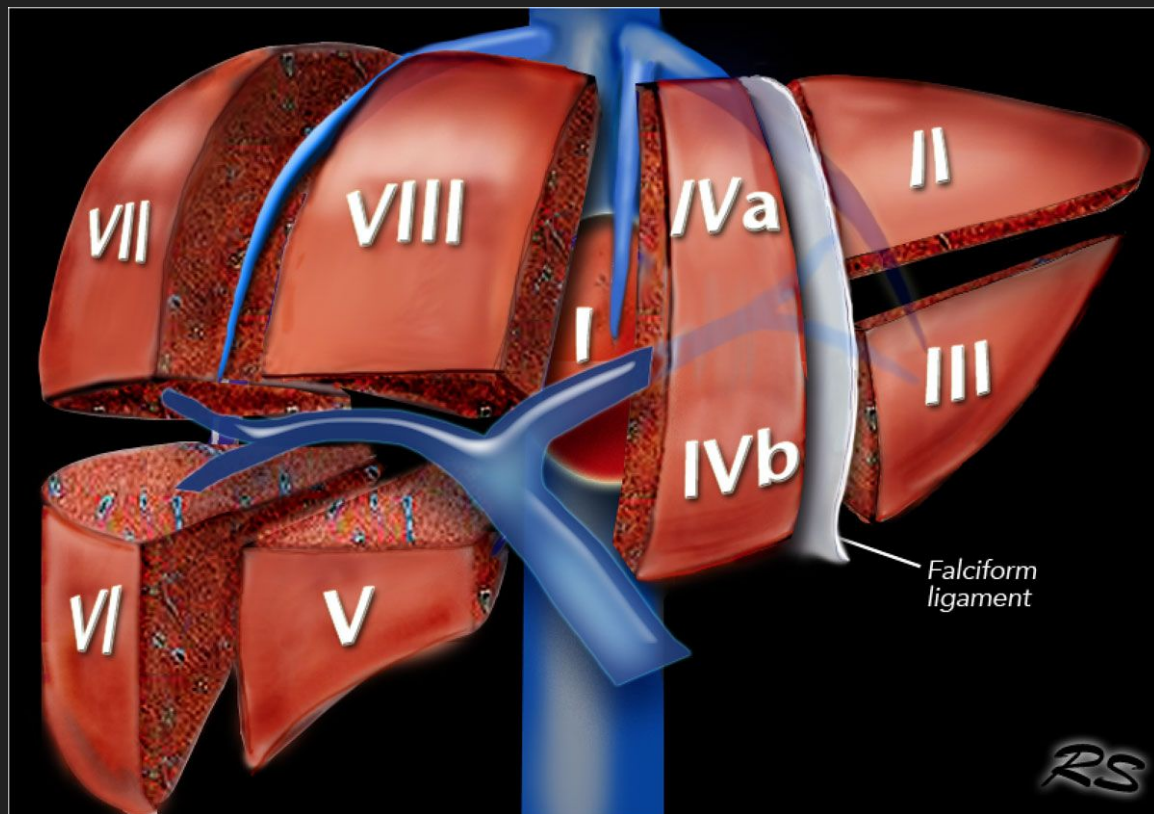
Intercostal

心中有解剖 什麼view都不重要





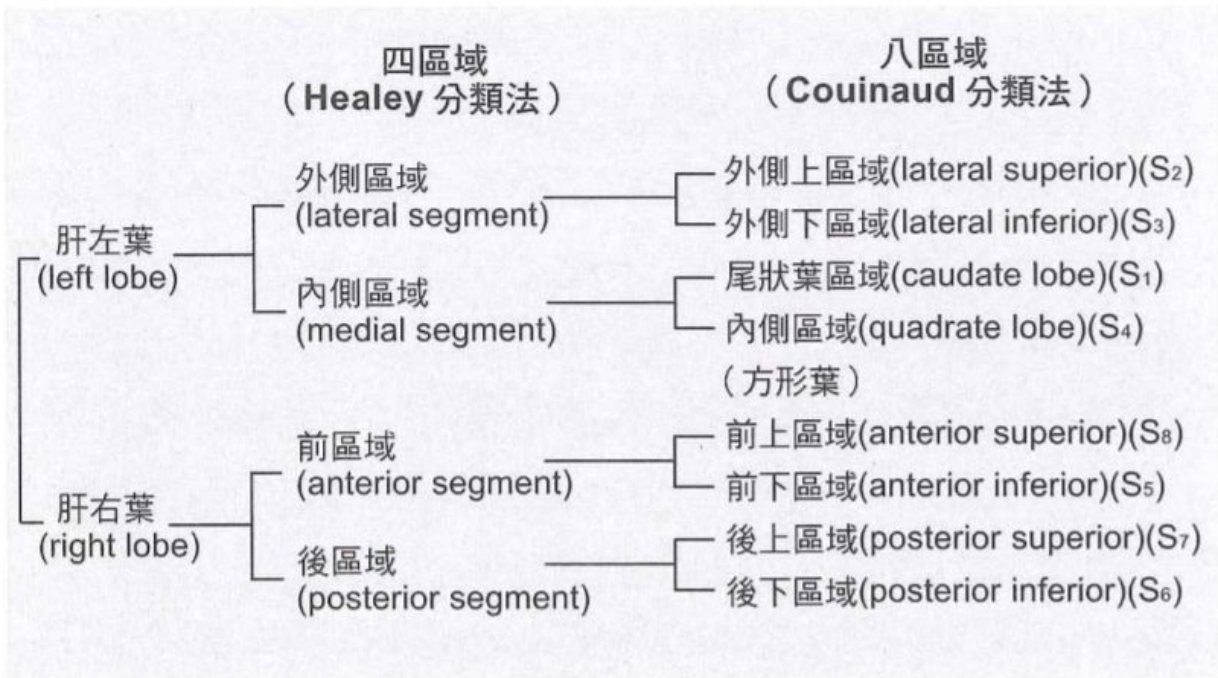
肝臟的基本解剖構造





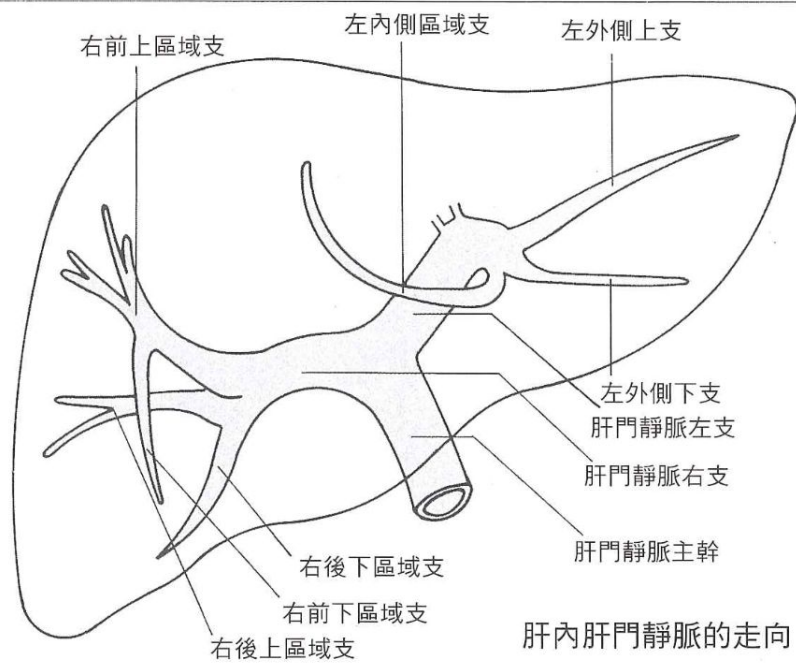
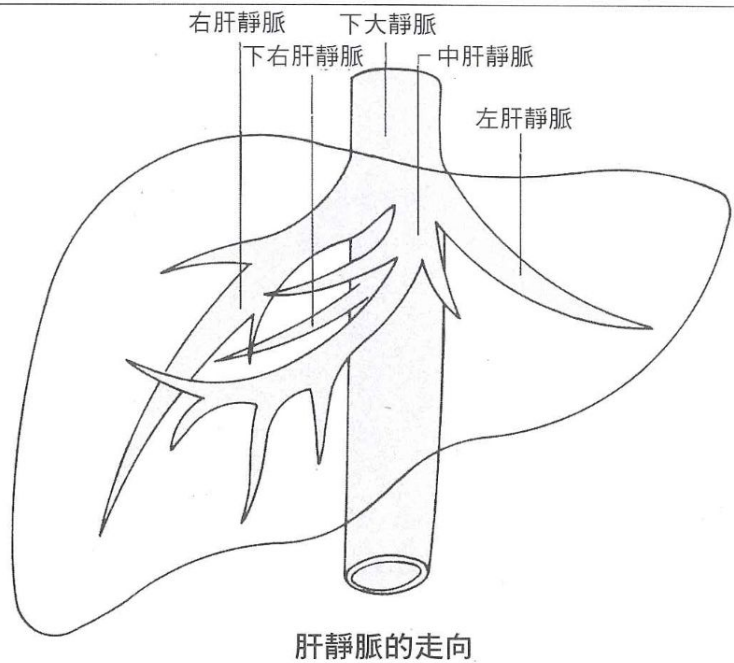
肝臟的基本解剖構造

肝的區域分類



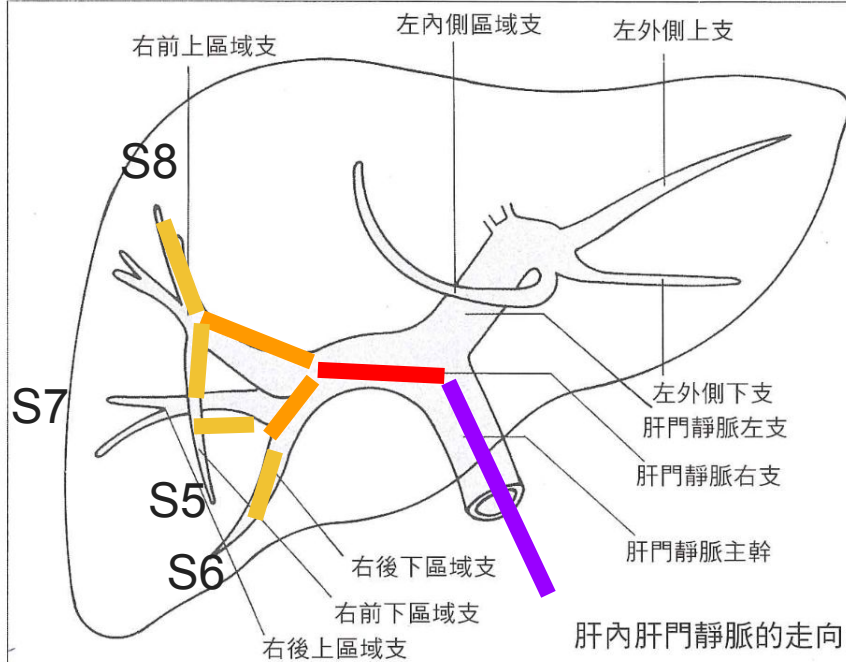
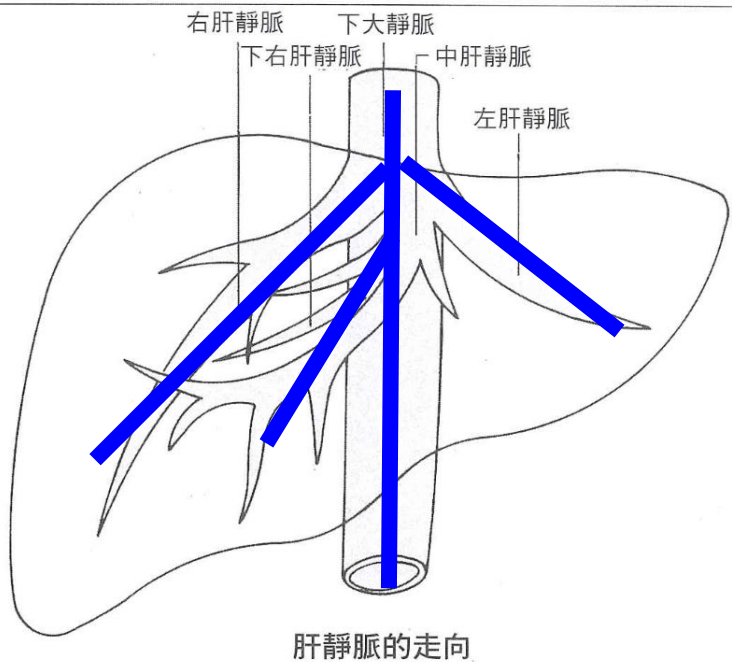


肝臟的血管走向



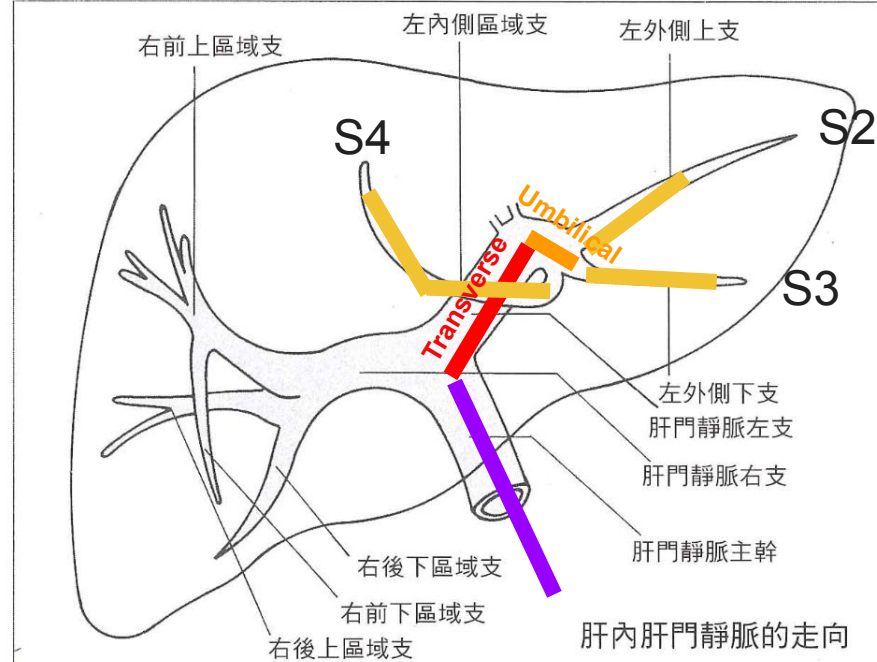
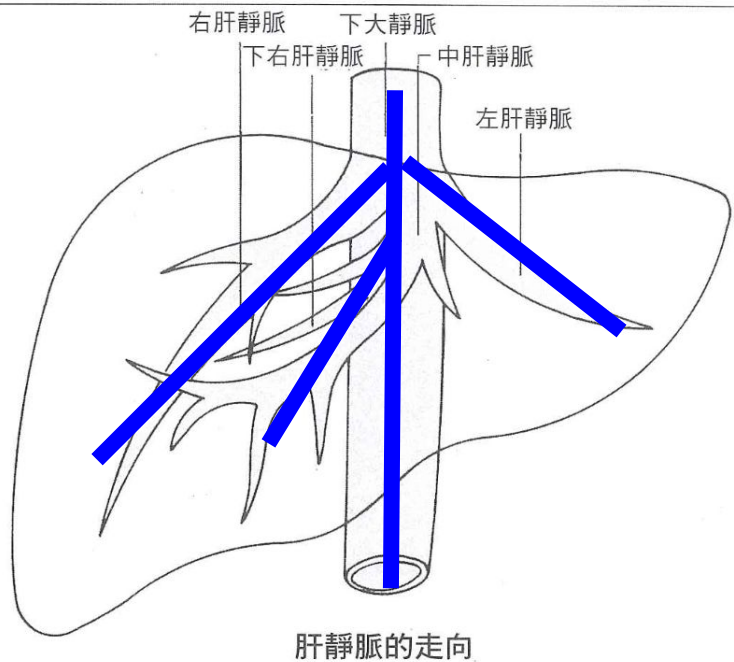


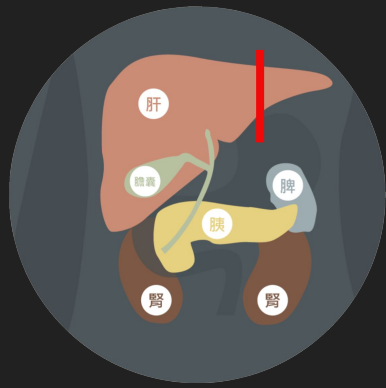
肝臟的血管走向



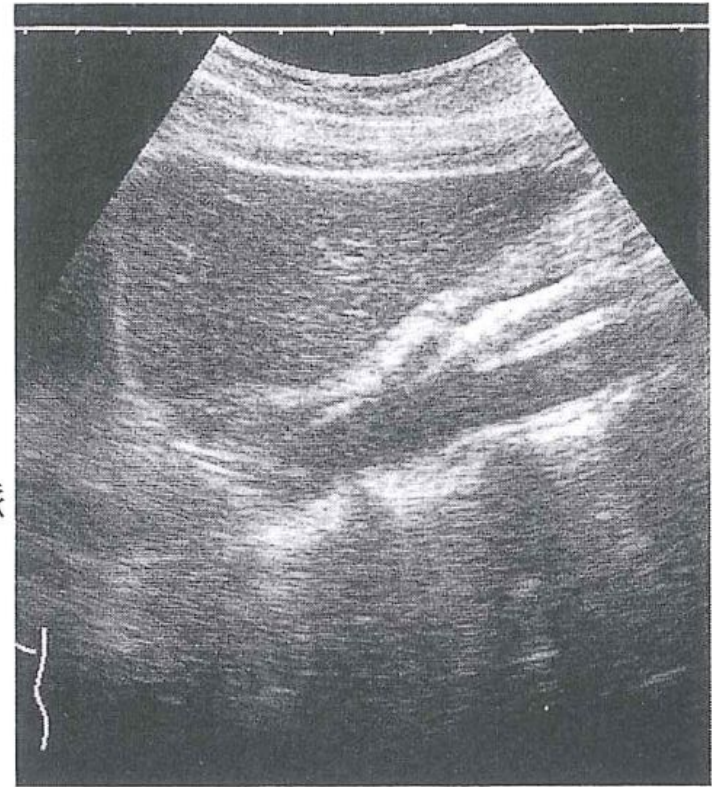
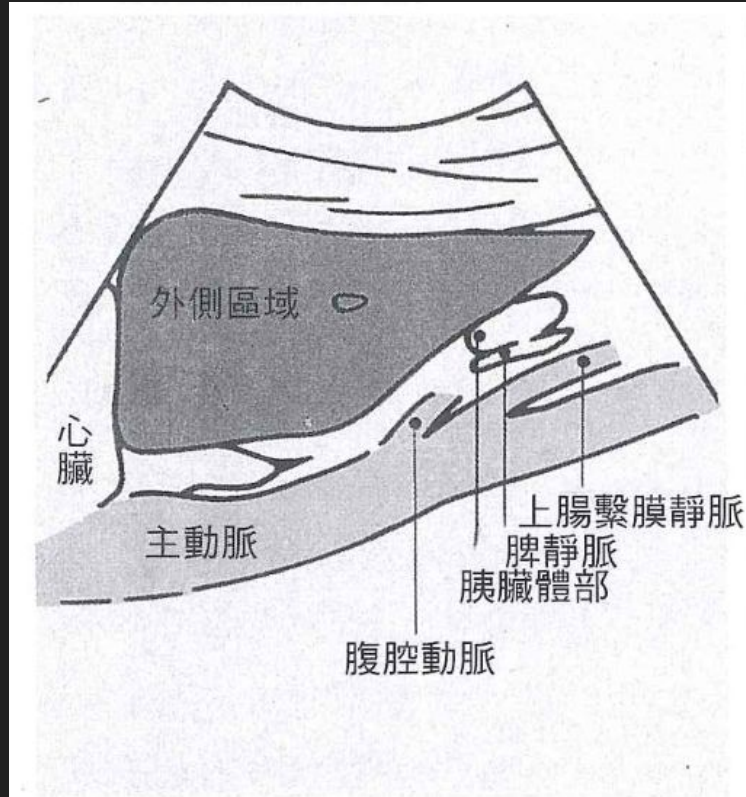


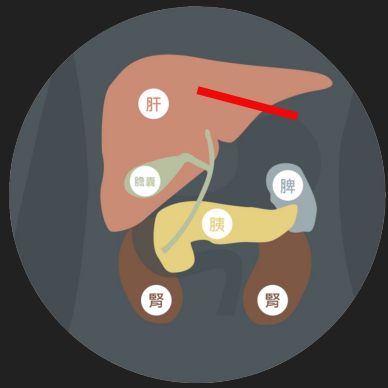
肝臟的血管走向



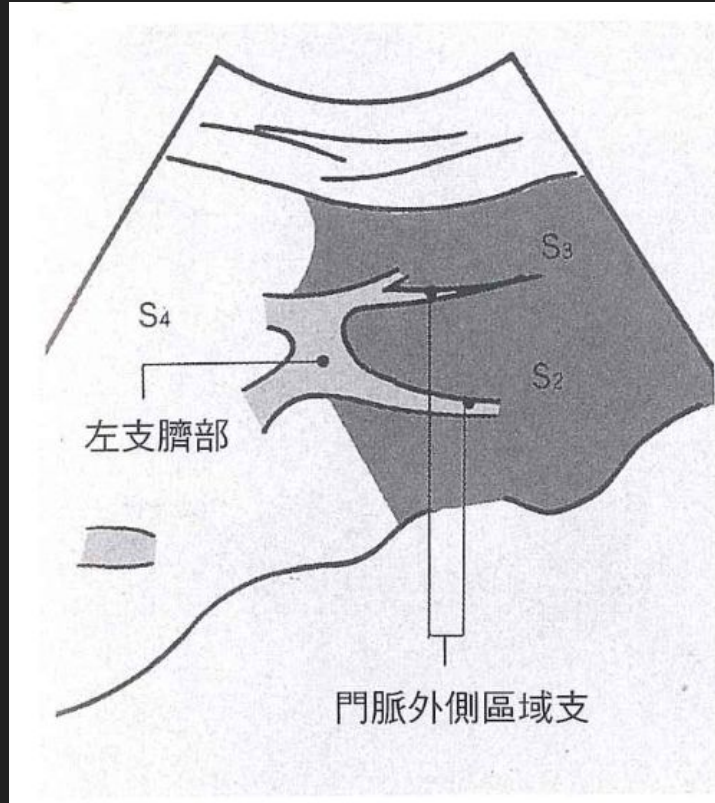


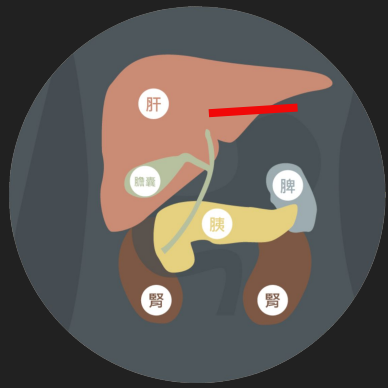
Lateral segment的各種view





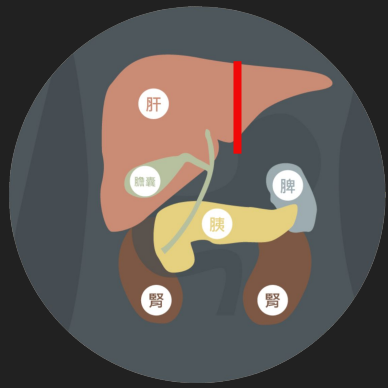
Lateral segment的各種view



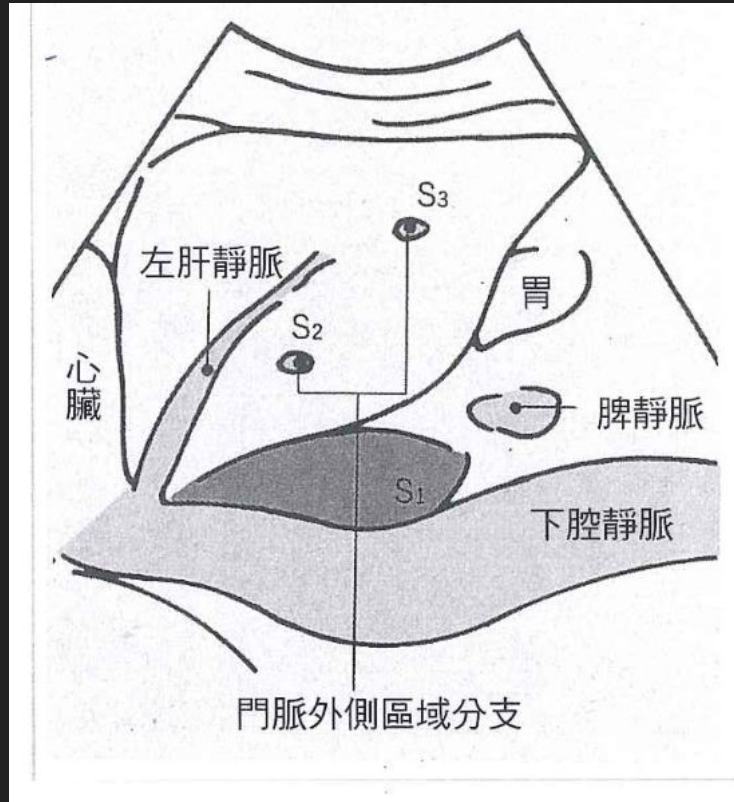


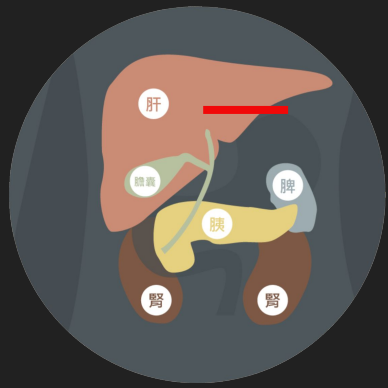
Lateral segment的各種view



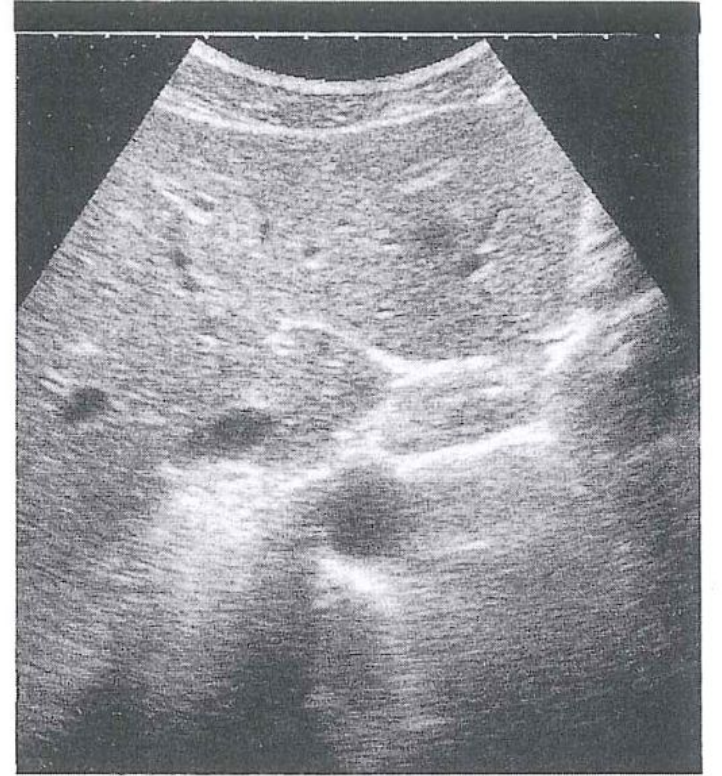
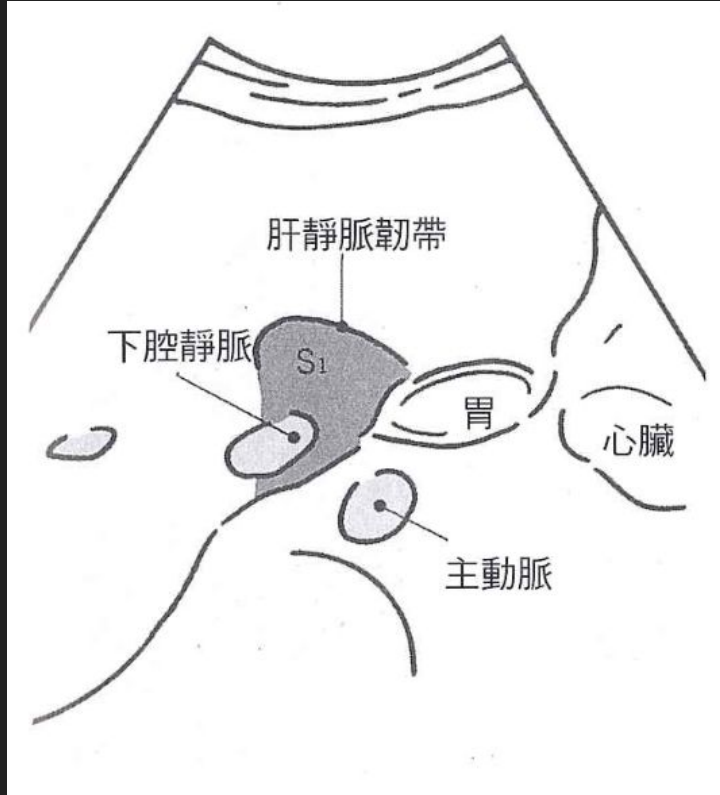


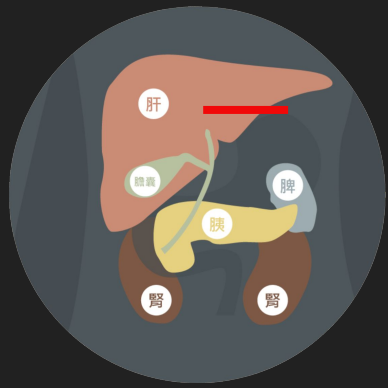
Caudate lobe的各種view



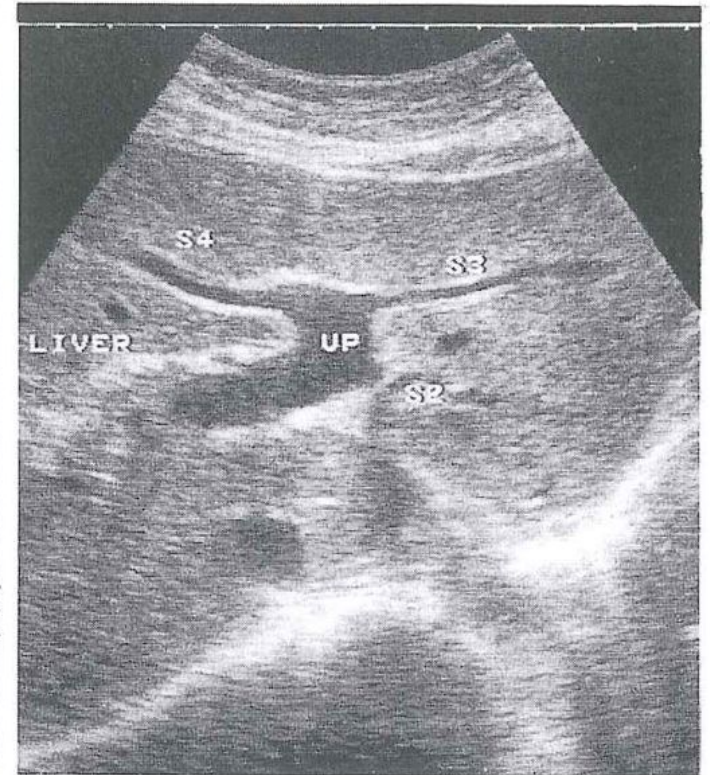
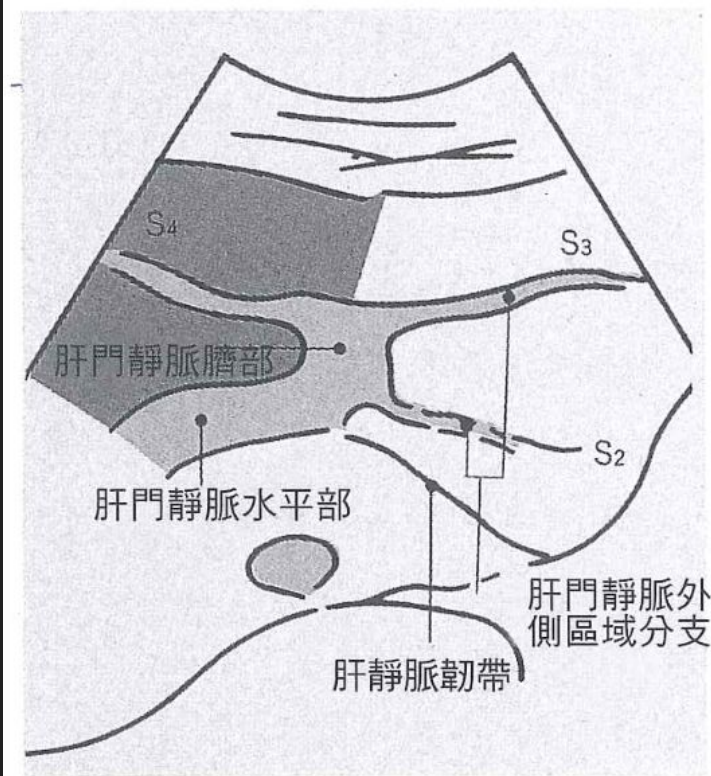


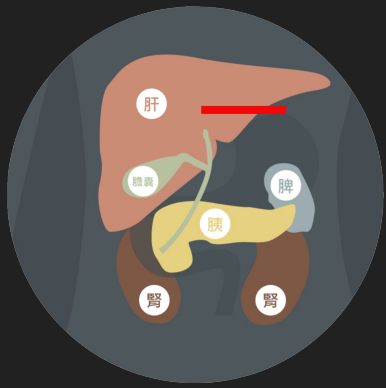
Caudate lobe的各種view



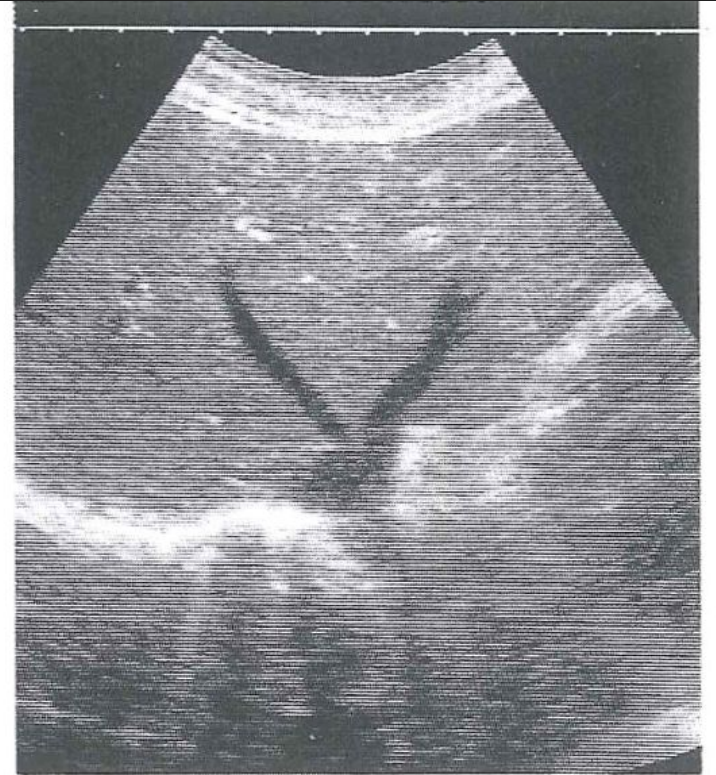
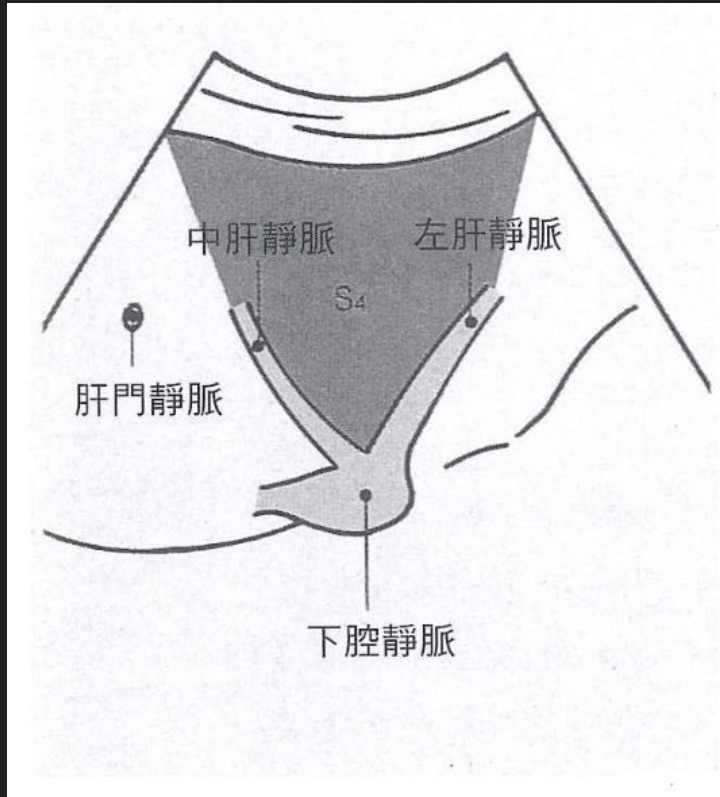


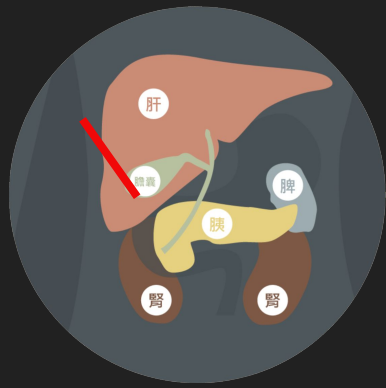
Medial Segment 的各種view



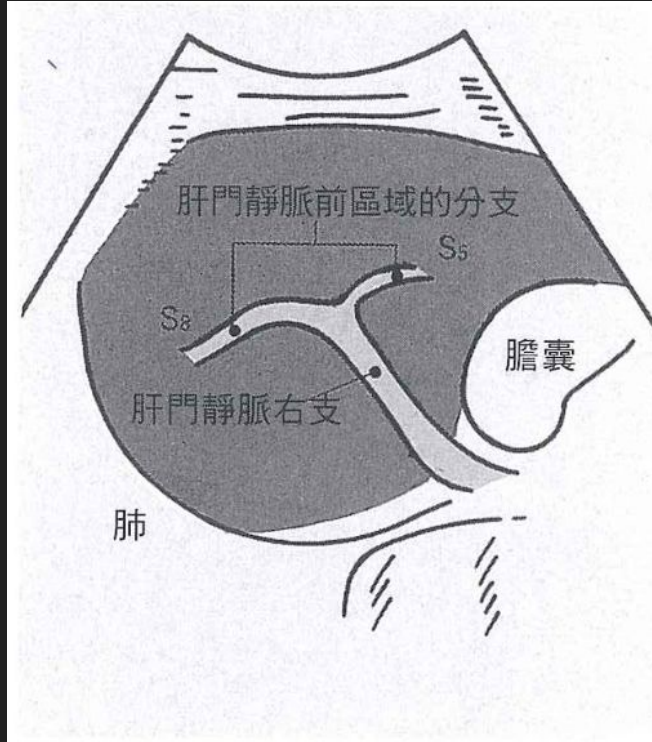


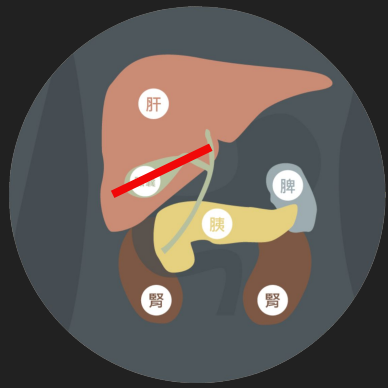
Medial Segment 的各種view



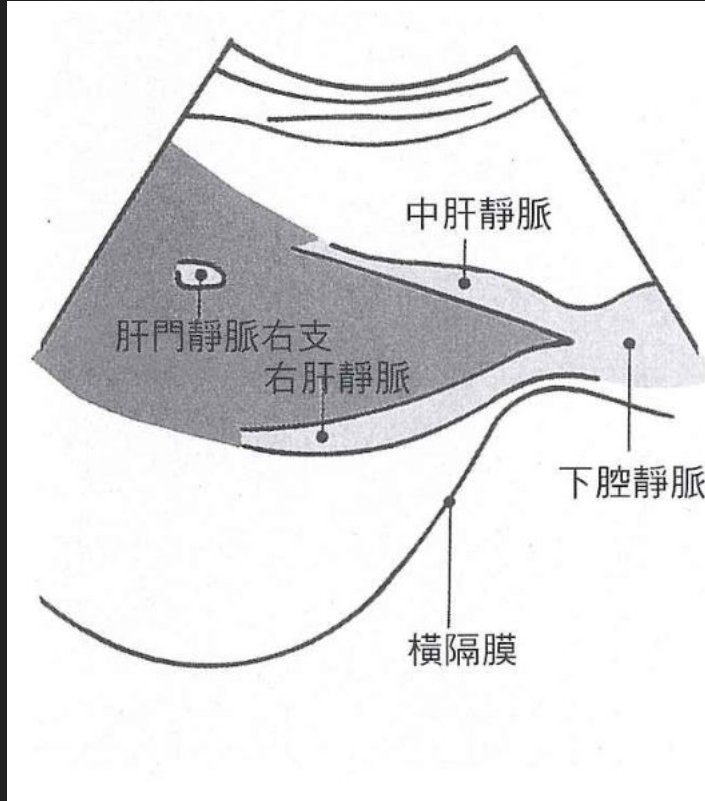


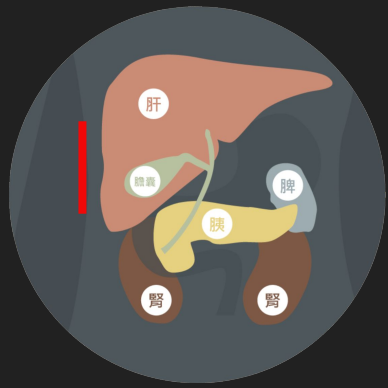
Anterior Segment 的各種view



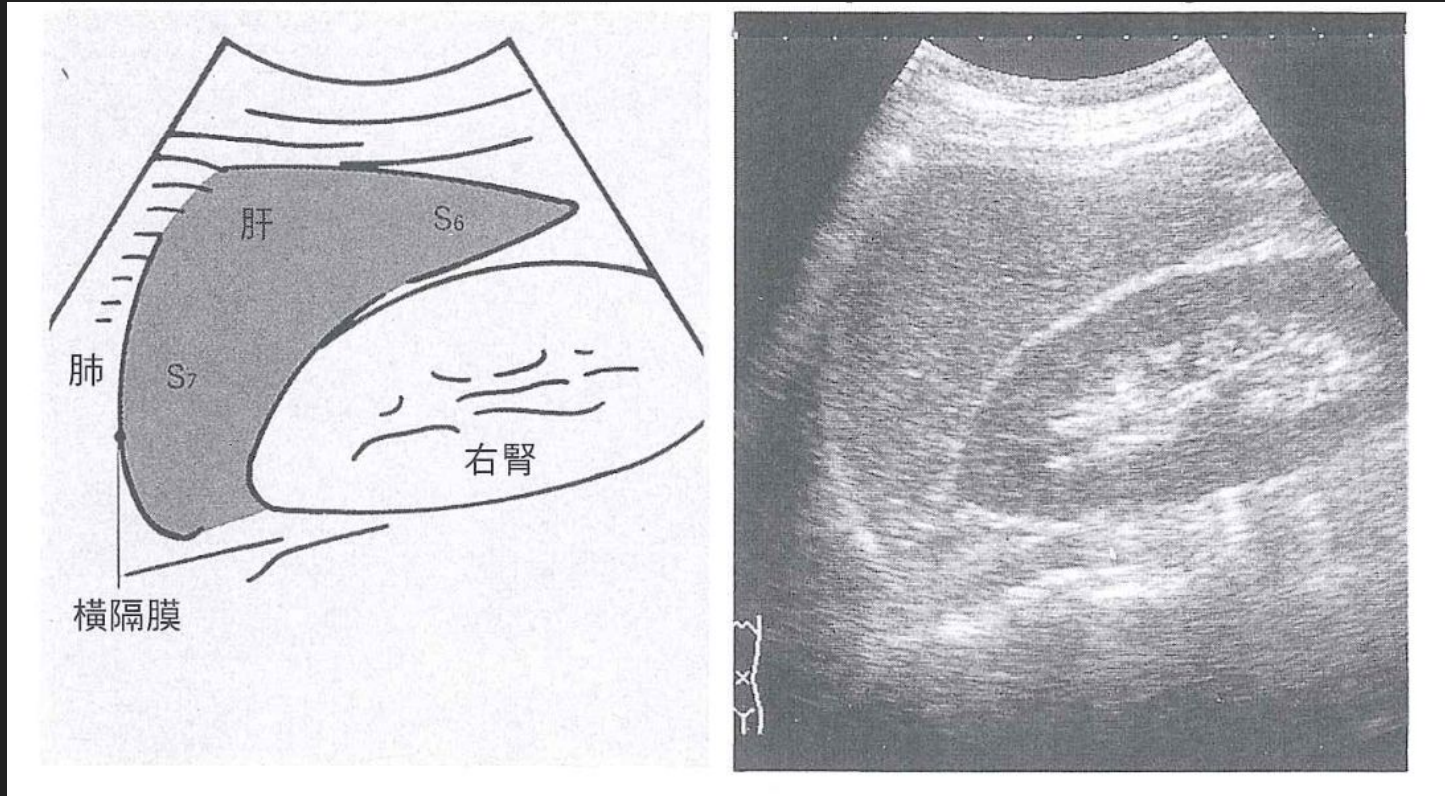


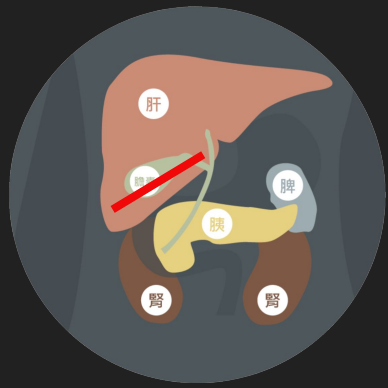
Anterior Segment 的各種view



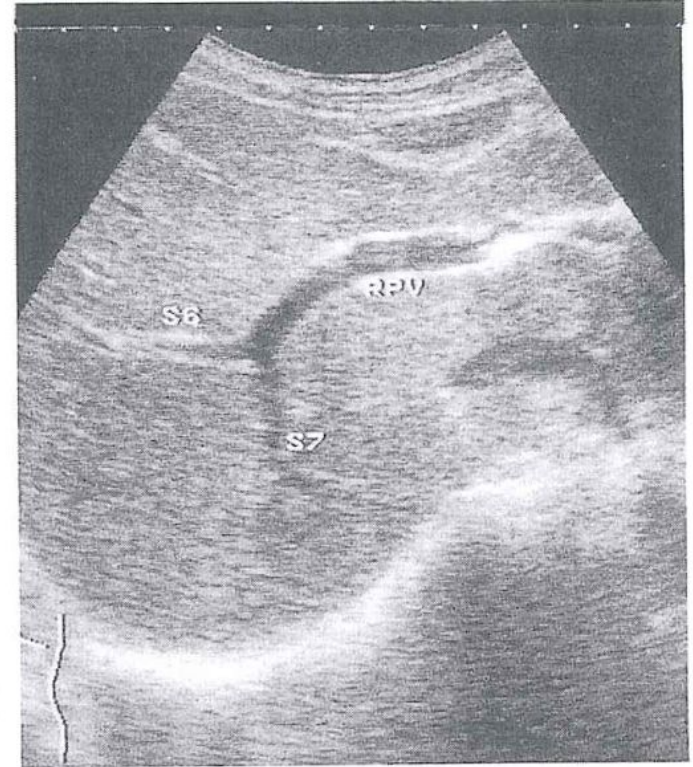
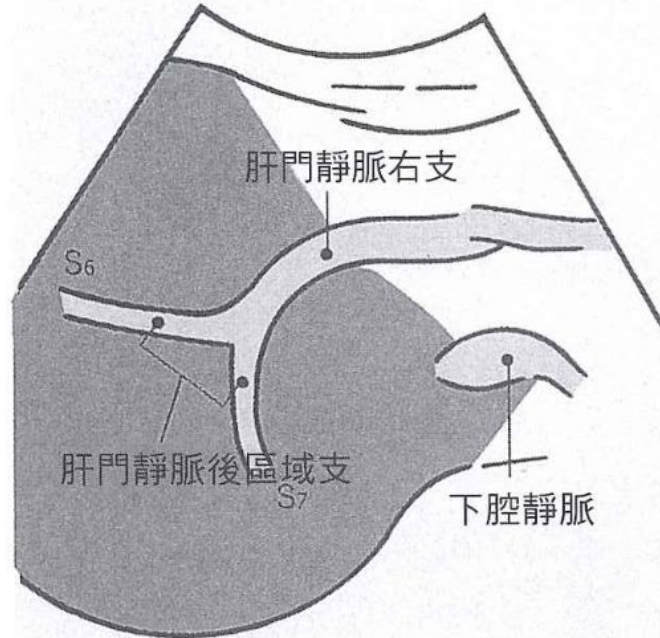


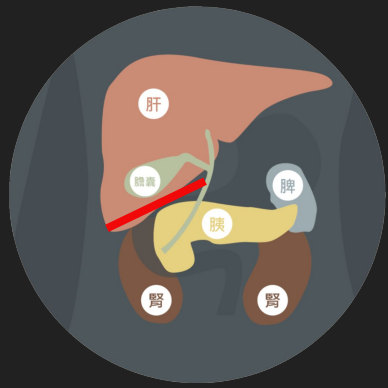
Posterior Segment 的各種view



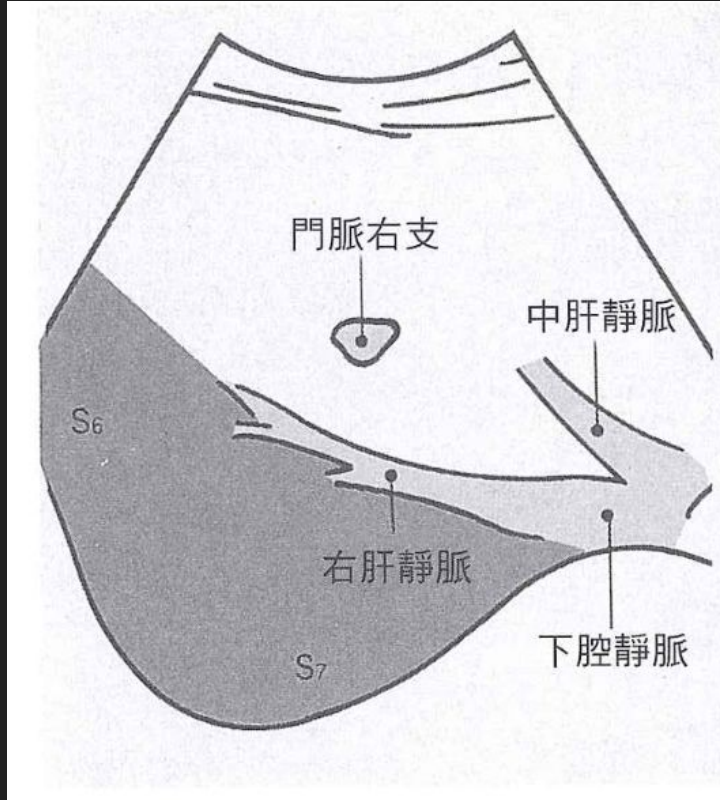


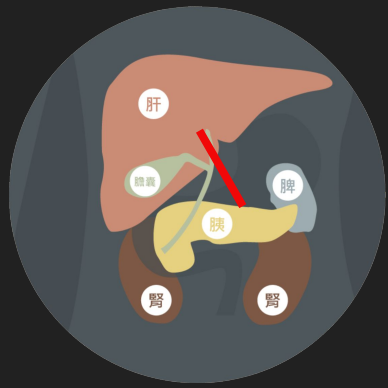
Posterior Segment 的各種view



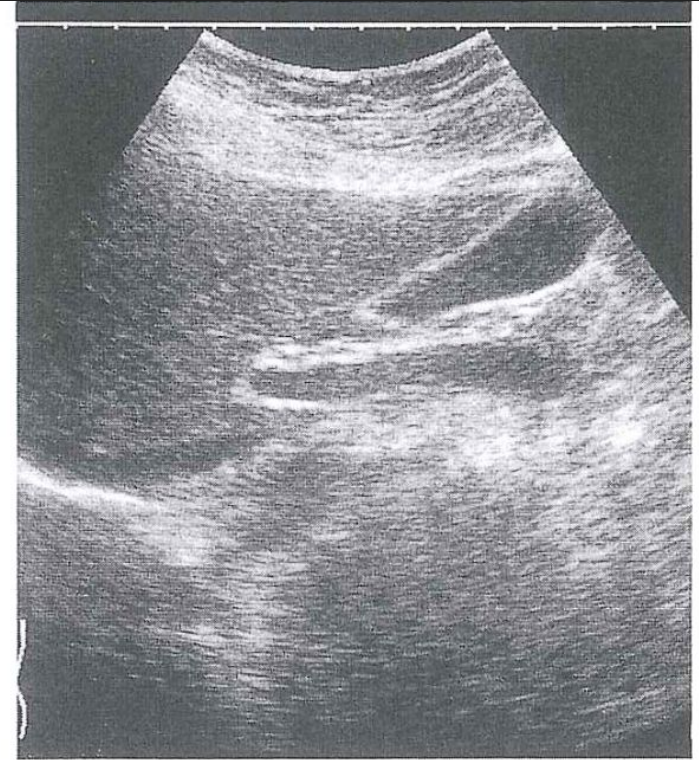
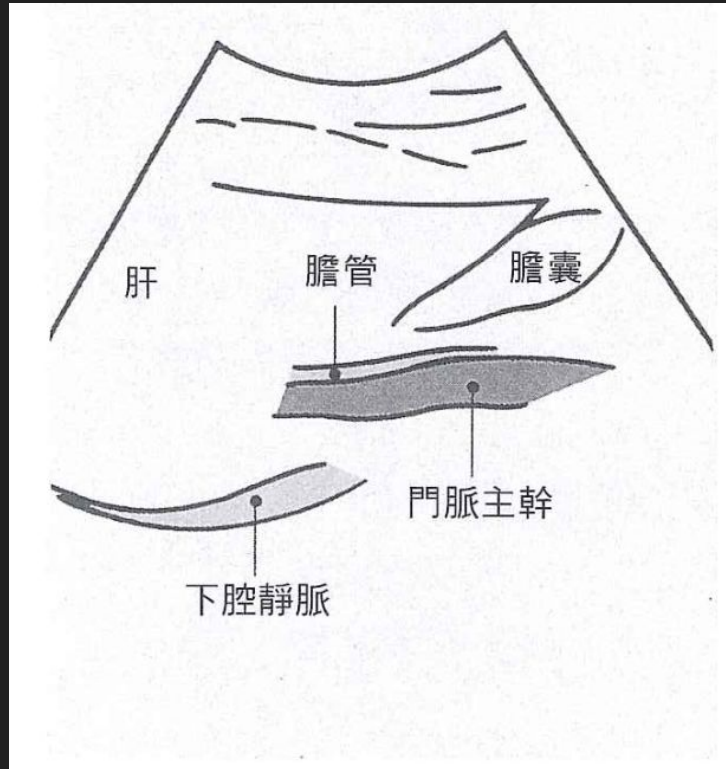


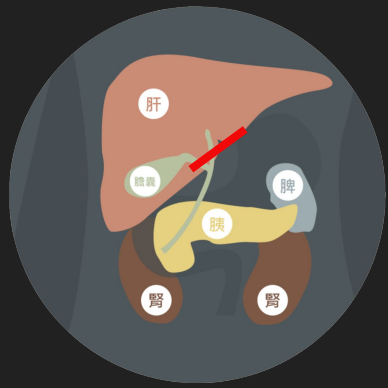
Posterior Segment 的各種view



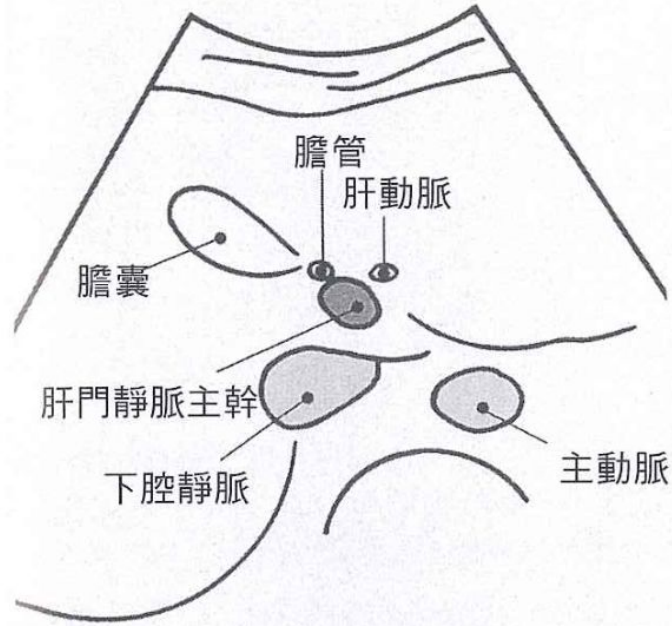


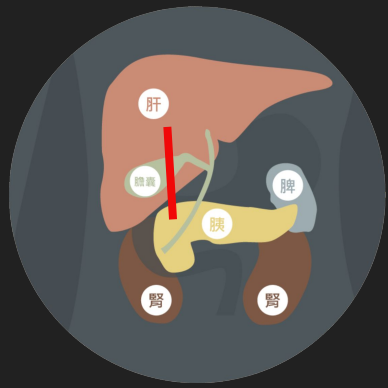
Portal Vein 的各種 view



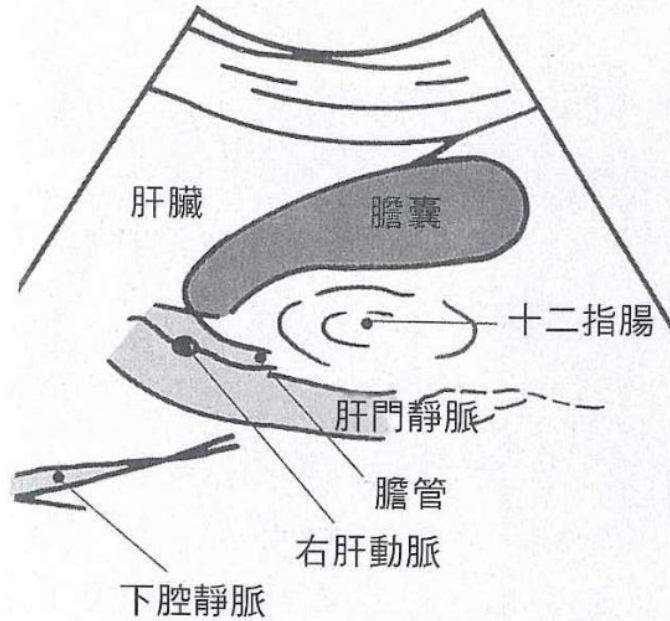


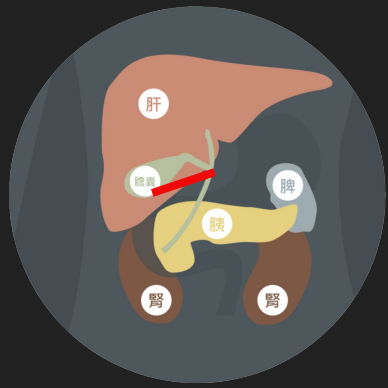
Portal Vein 的各種 view



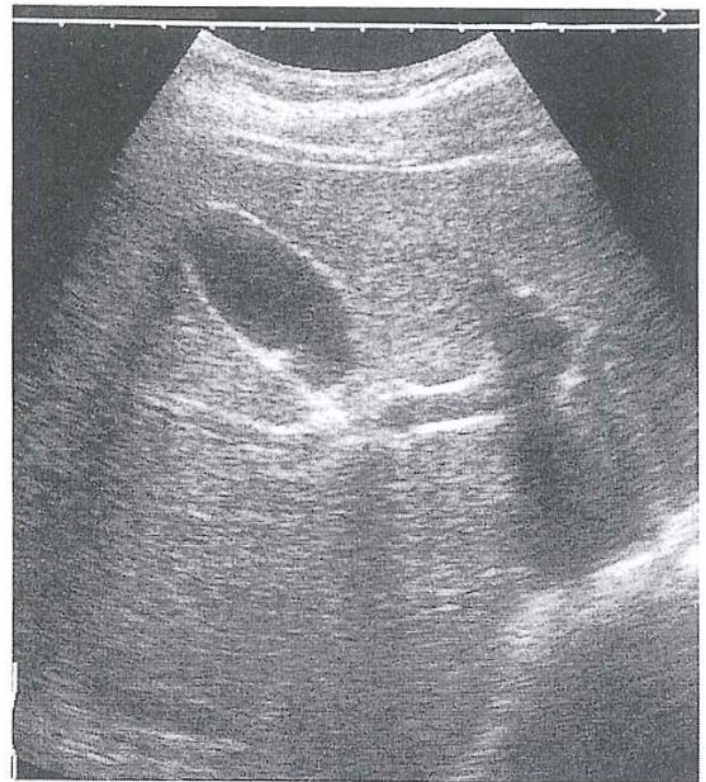
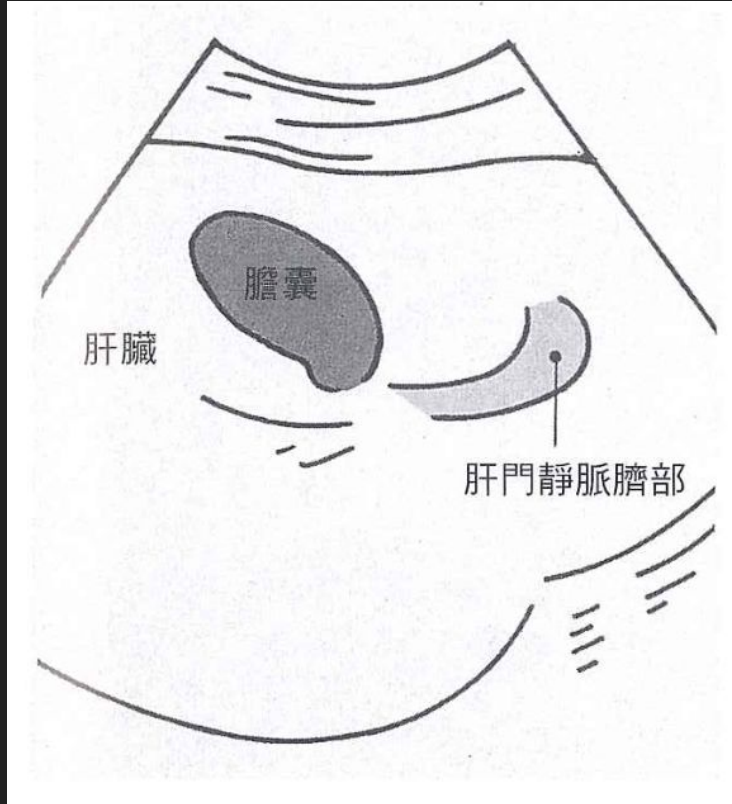


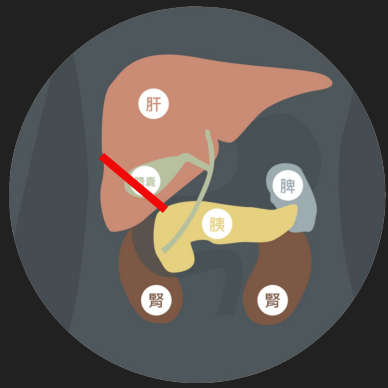
Gallbladder 的各種view



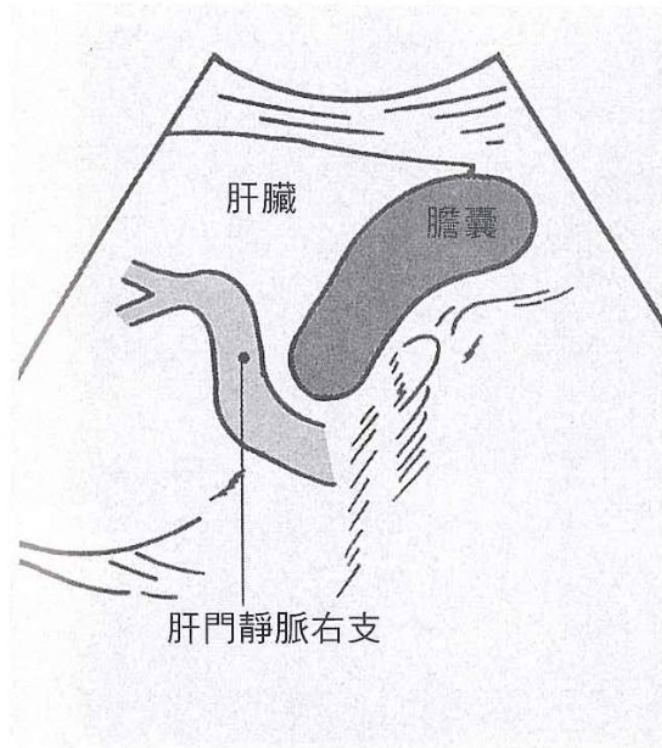


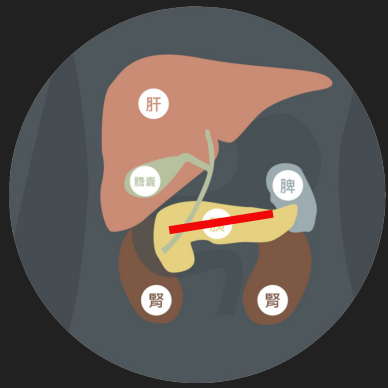
Gallbladder 的各種view



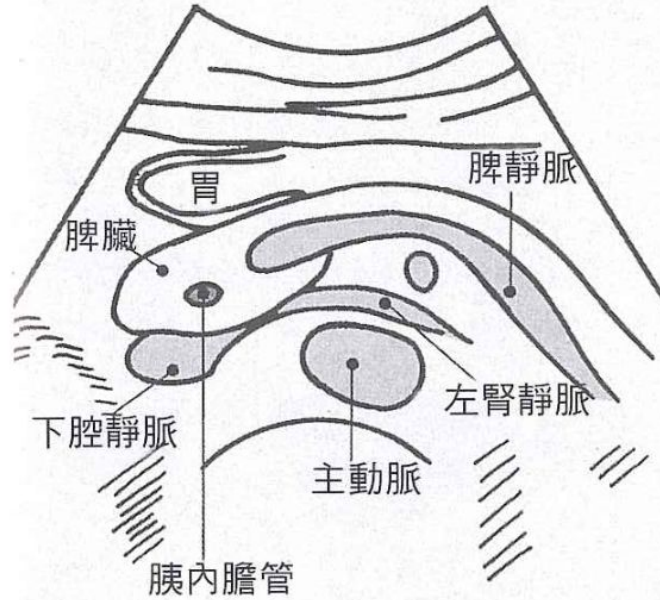


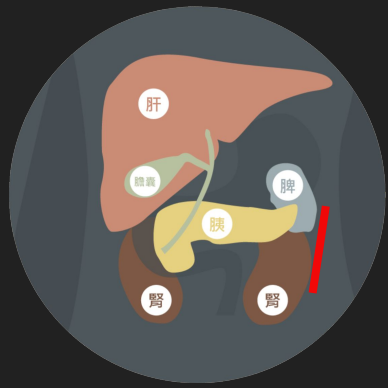
Gallbladder 的各種view



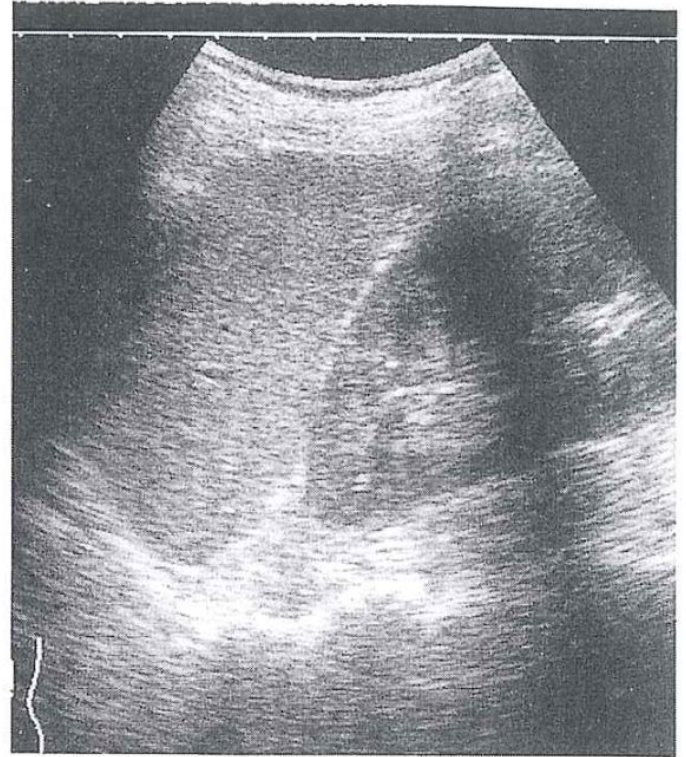
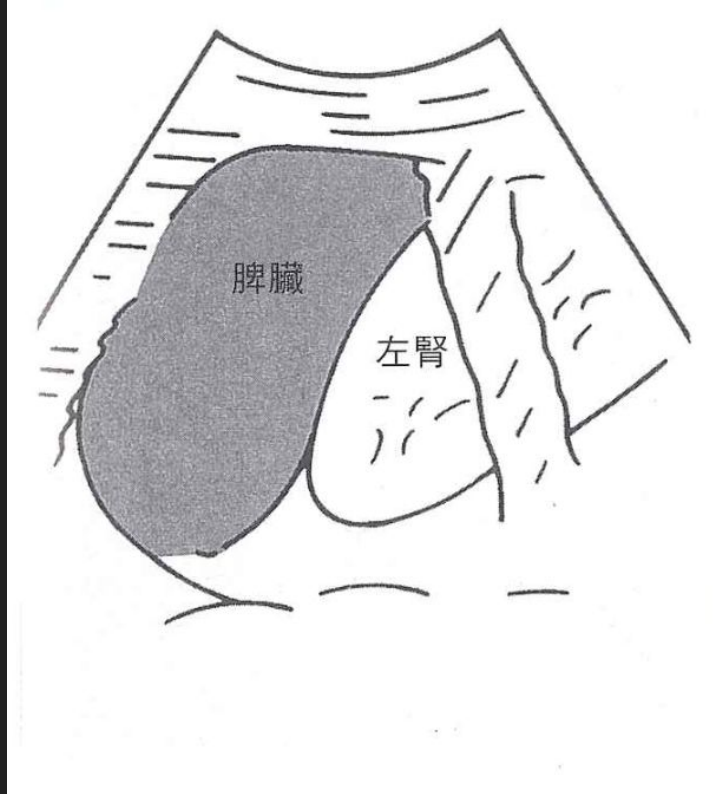


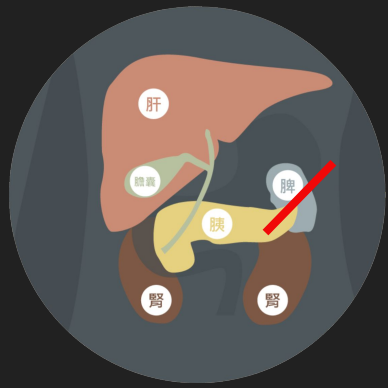
Pancreas 的各種 view



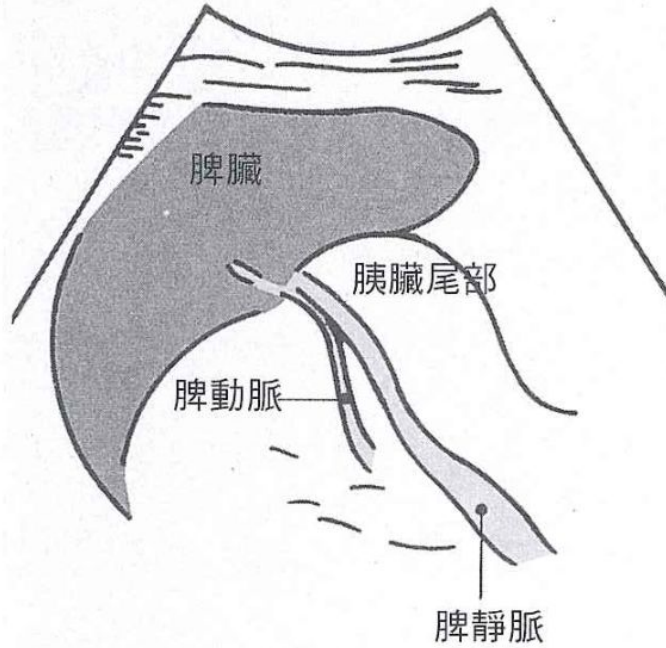


Spleen 的各種view

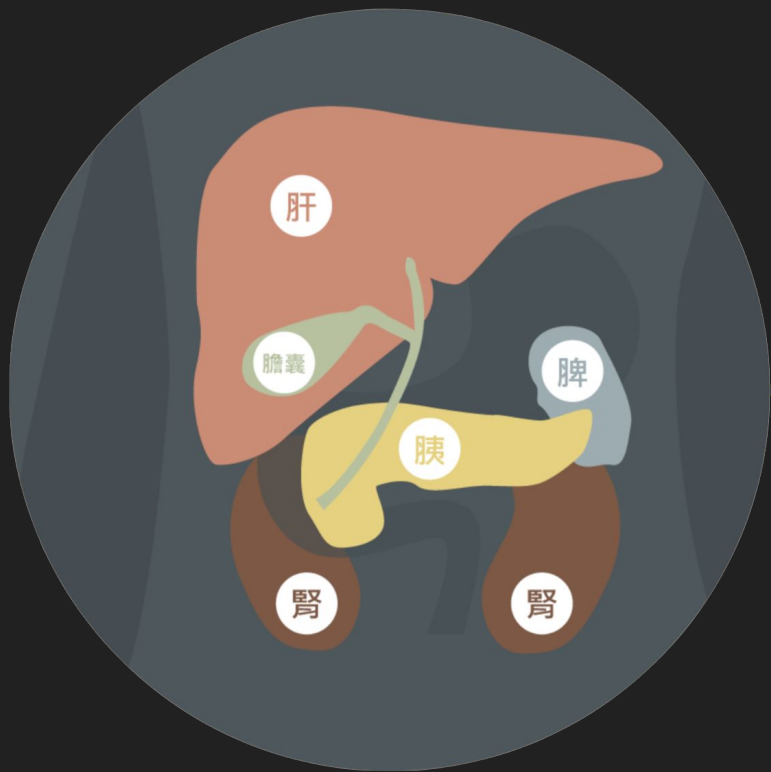




Spleen 的各種view



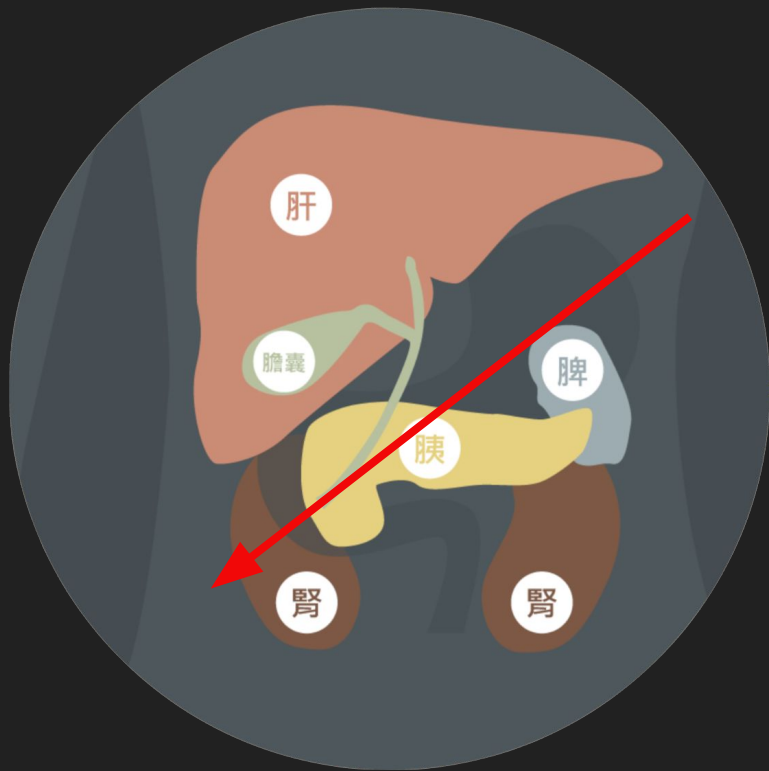
綜合來說：每個部位都要兩種 view 掃過



綜合來說：每個部位都要兩種 view 掃過

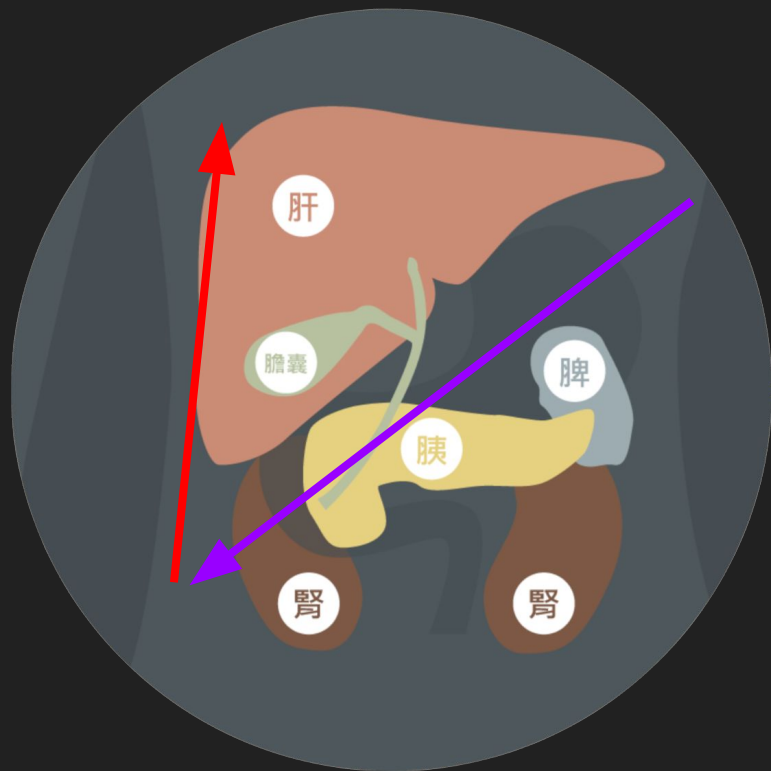
8字攻略

1. Sagital Lt → Rt





綜合來說：每個部位都要兩種 view 掃過



8字攻略

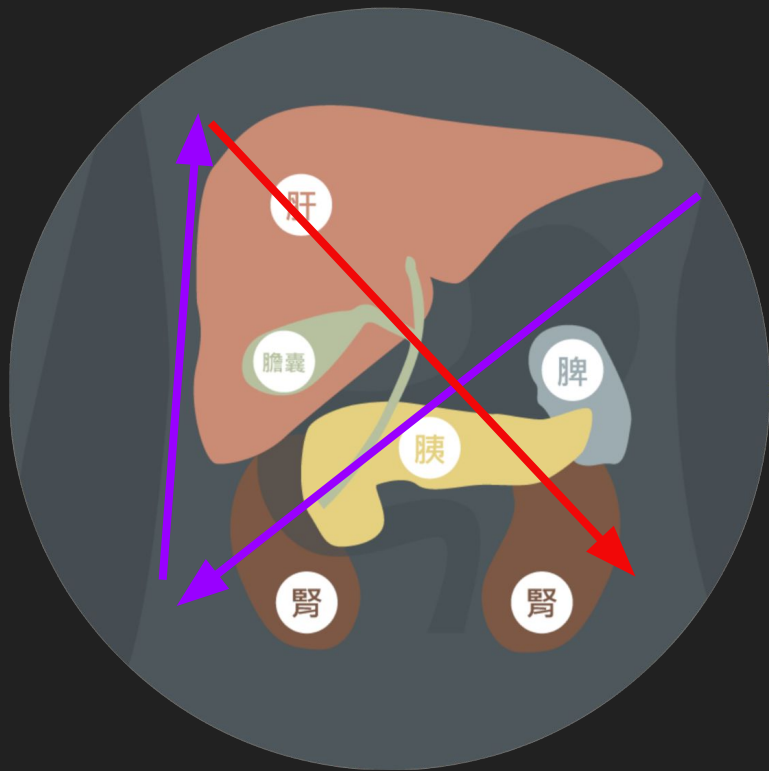
1. Sagital Lt → Rt
2. Intercostal Rt



綜合來說：每個部位都要兩種 view 掃過

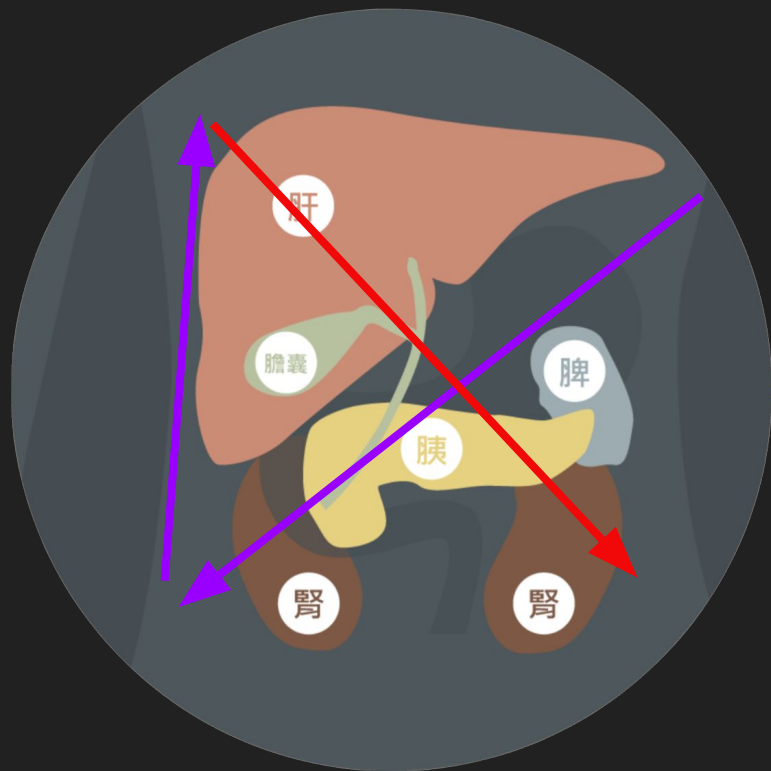
8字攻略

1. Sagital Lt → Rt
2. Intercostal Rt
3. **Subchondral Rt → Lt**





綜合來說：每個部位都要兩種 view 掃過

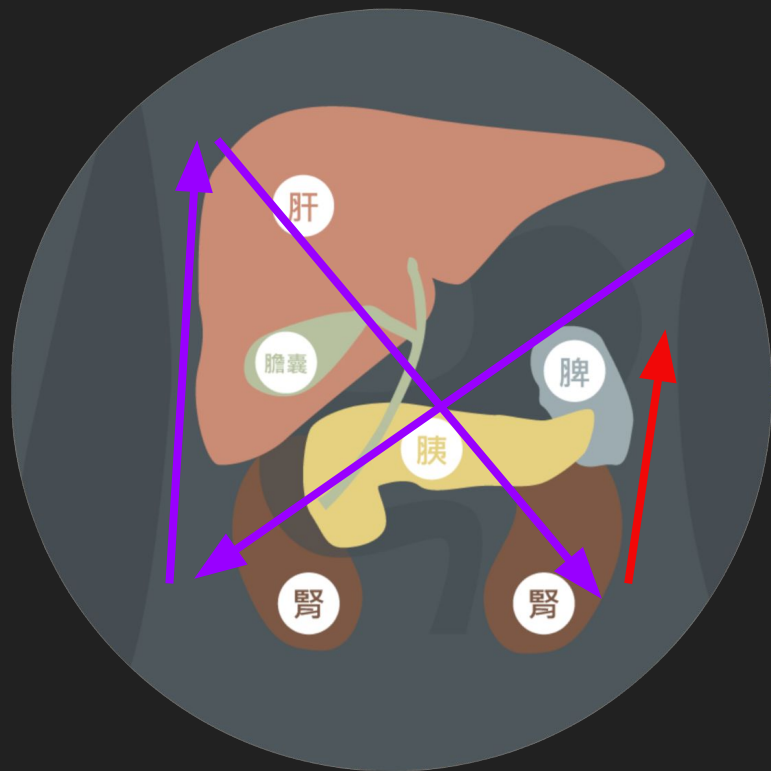


8字攻略

1. Sagital Lt → Rt
2. Intercostal Rt
3. Subchondral Rt → Lt
4. **Transverse**



綜合來說：每個部位都要兩種 view 掃過



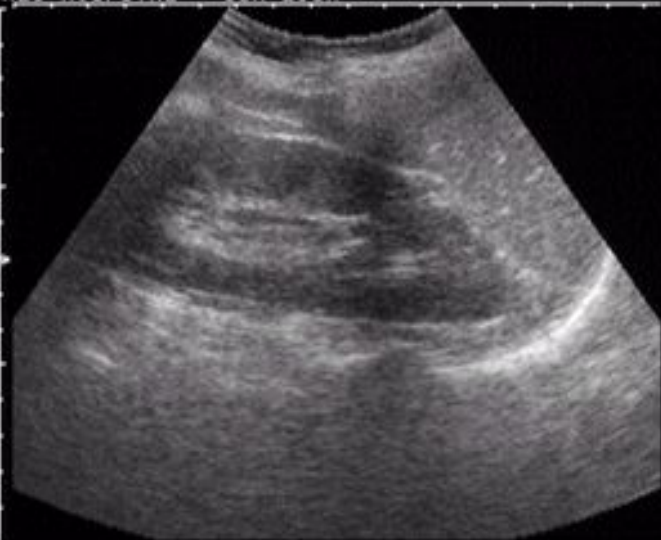
8字攻略

1. Sagital Lt → Rt
2. Intercostal Rt
3. Subchondral Rt → Lt
4. Transverse
5. Lt side sagital
6. Lt side intercostal

17991304
I-HEI HOSPITAL SSA-250A

PMR1R
< 4 > 12-25.04
09:52:44

C3.75
18HZ



R/L/L
RD/ 78
14. KCM

17991304
I-HEI HOSPITAL SSA-250A

PMR1R
< 4 > 12-25.04
09:52:25

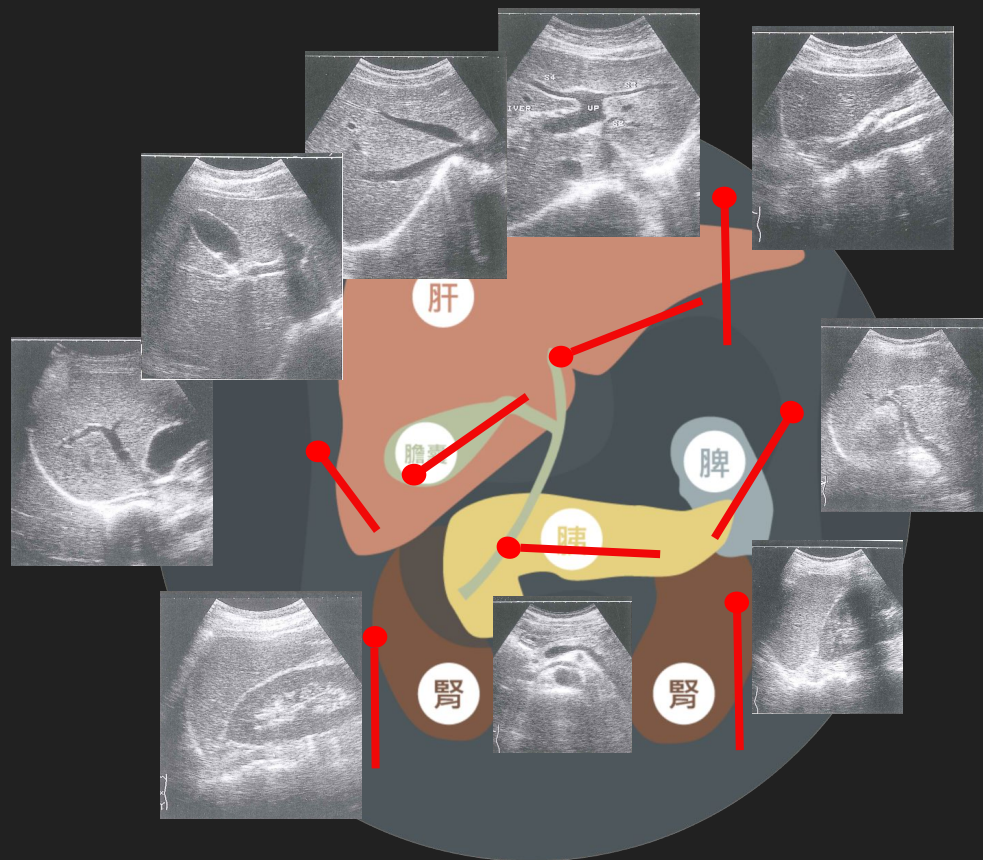
C3.75
18HZ



R/L/L
RD/ 78
14. KCM

必掃清單

- Liver edge
- 工字型(左肝)
- Hepatic v
- Portal triad
- 膽囊
- 腎臟
- 脾臟
- 胰臟



第二篇章：值班除了切 CT 還能怎麼辦

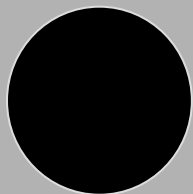
Fever

Abdominal distention

Abdominal tenderness

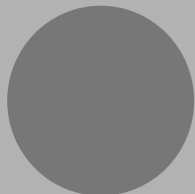
GI bleeding

超音波的大原則



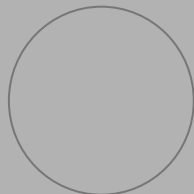
無回音型
Anechoic

cyst
vessel
fluid



低回音型
Hypoechoic

tumor
abscess (異質)



同回音型
Isoechoic



高回音型
Hyperechoic

air
fat
骨頭, 鈣化

發燒 - 都叫 IAI 就對了啦

- **Abscess**
 - Liver, Spleen, Kidney, Enteric
- **Biliary tract infection**
 - Cholecystitis, CBD stone cholangitis
- **Infected fluid accumulation**
 - Peritonitis, Pseudocyst, Infected cyst

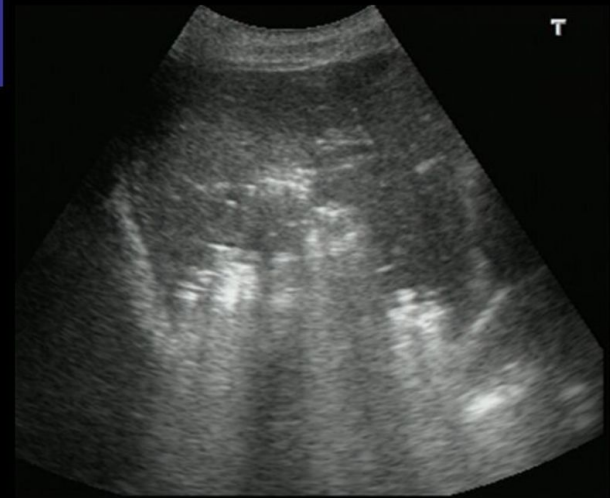
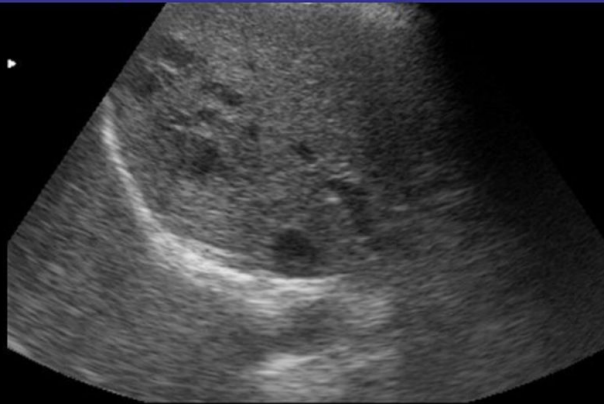
發燒 - 都叫 IAI 就對了啦

- **Abscess**
 - Liver, Spleen, Kidney, Enteric
- **Biliary tract infection**
 - Cholecystitis, CBD stone cholangitis
- **Infected fluid accumulation**
 - Peritonitis, Pseudocyst, Infected cyst

Hepatic abscess

Abdominal pain + sepsis

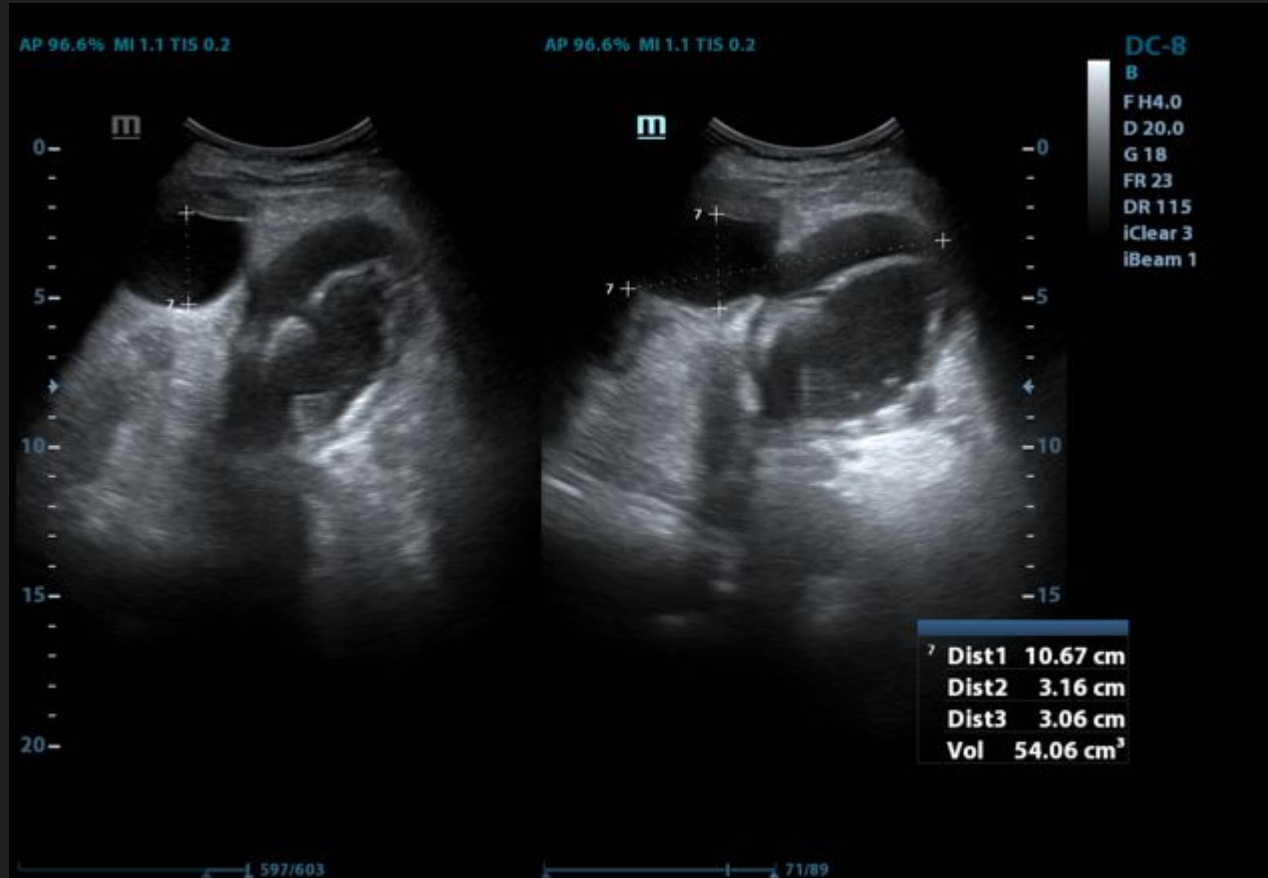
- 1 邊緣凹凸不平
- 2 壁肥厚
- 3 內部回音細微，不明顯（壞死物質）
- 4 有時可看到高回音部及伴隨在後方的音陰（因氣體而造成）
- 5 後方回音增強
- 6 與周圍的界線不清楚



Abscess everywhere



Biloma



發燒 - 都叫 IAI 就對了啦

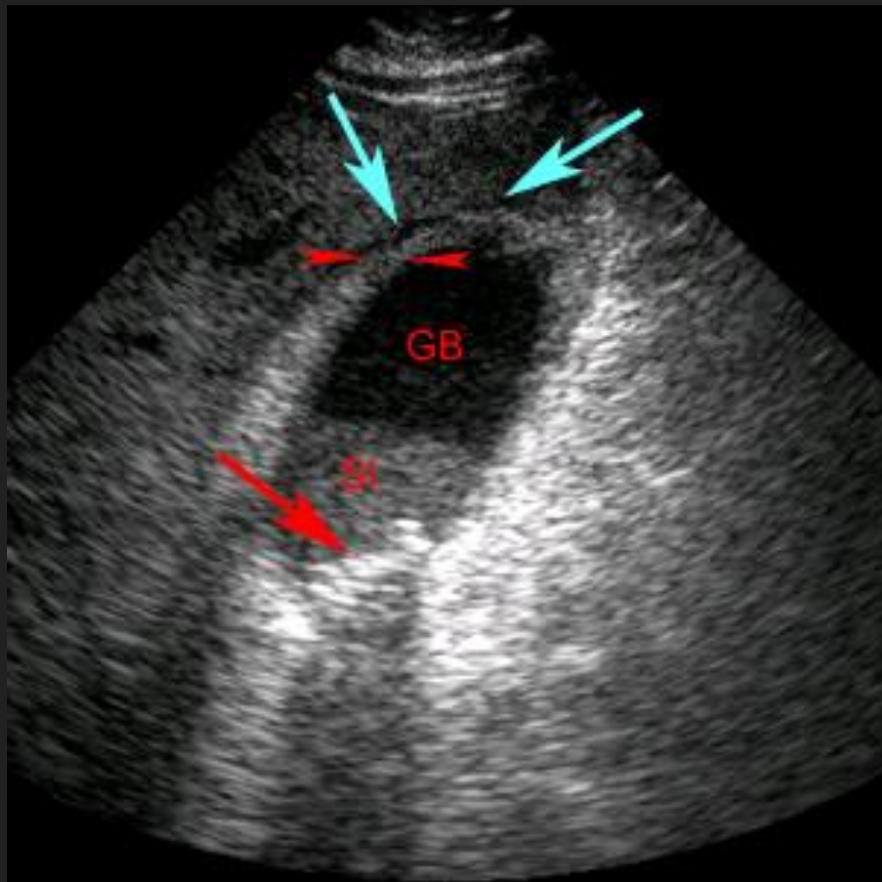
- Abscess
 - Liver, Spleen, Kidney, Enteric
- **Biliary tract infection**
 - Cholecystitis, CBD stone cholangitis
- Infected fluid accumulation
 - Peritonitis, Pseudocyst, Infected cyst

Acute Cholecystitis

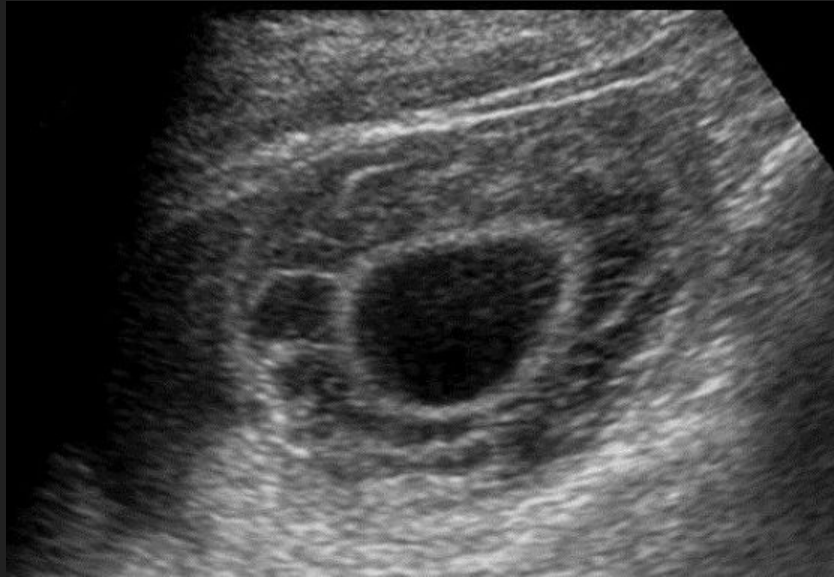
- 1 膽囊腫大 (8 × 3cm 以上)
- 2 膽囊壁肥厚 (3mm 以上)
- 3 回音透明帶*
- 4 膽泥積留
- 5 膽囊頸部有結石嵌頓
- 6 膽囊周圍有液體貯留

* 膽囊黏膜和漿膜以高回音顯像，兩層之間呈低回音的透明帶，因此膽囊壁看起來像有三層結構，這又稱為 double wall sign。

7. Echo murphy sign



Not all GB thickening is cholecystitis



- Portal Hypertension (LC)
- Congestive heart failure
- Hepatitis
- Hypoalbuminemia
- Ascites



Congestive liver

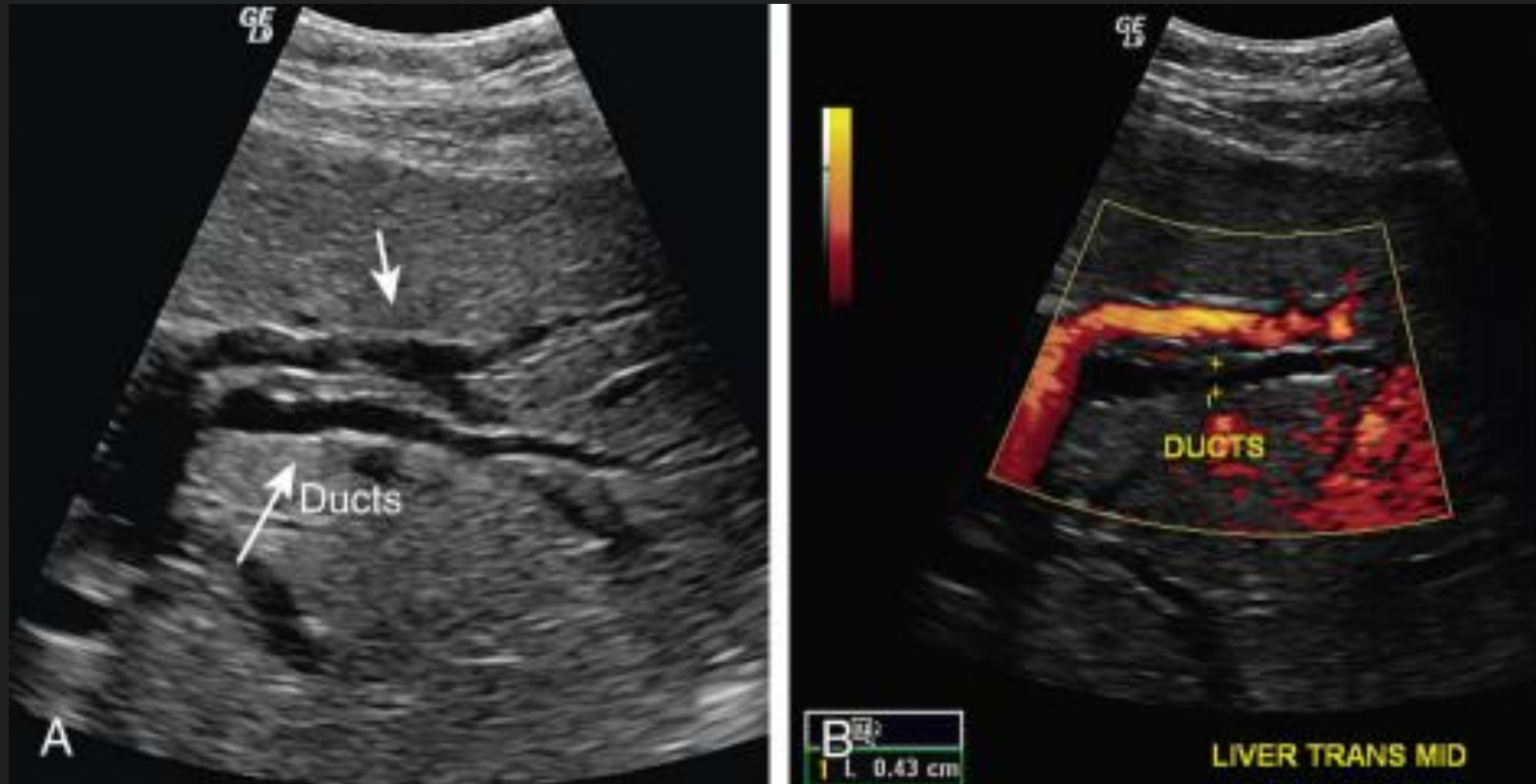


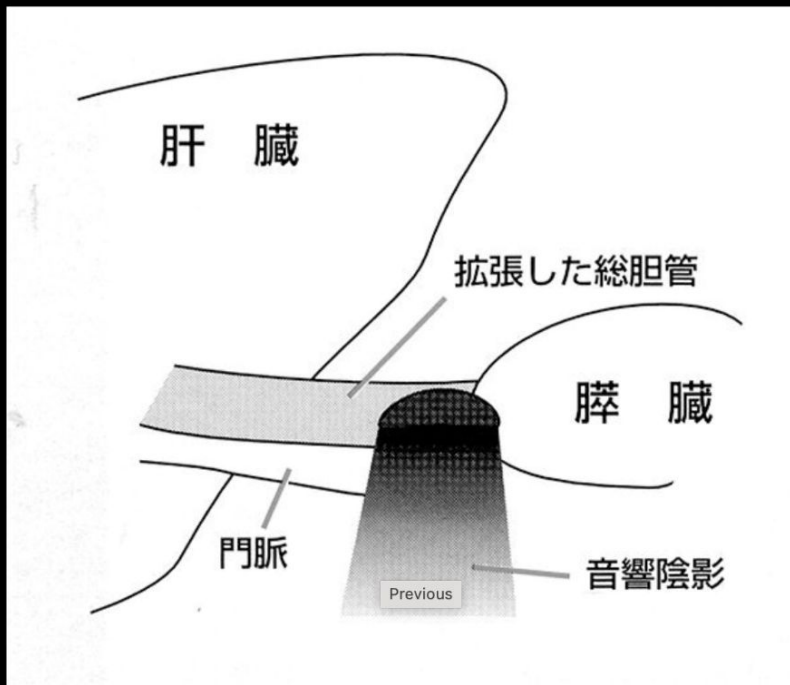
Shotgun Sign



7-11 rule

IHD dilatation: Parallel channel sign





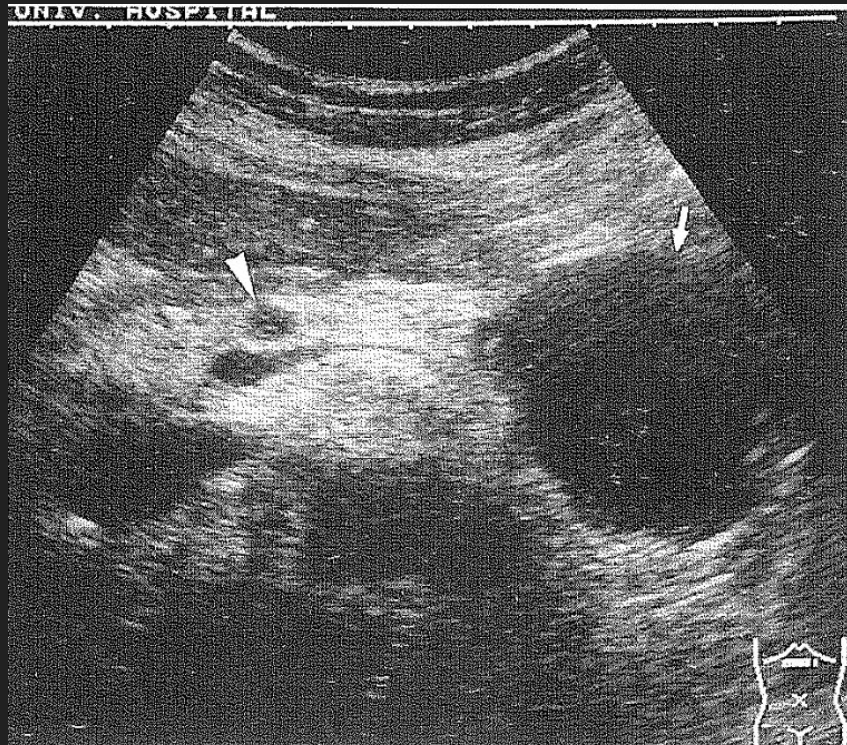
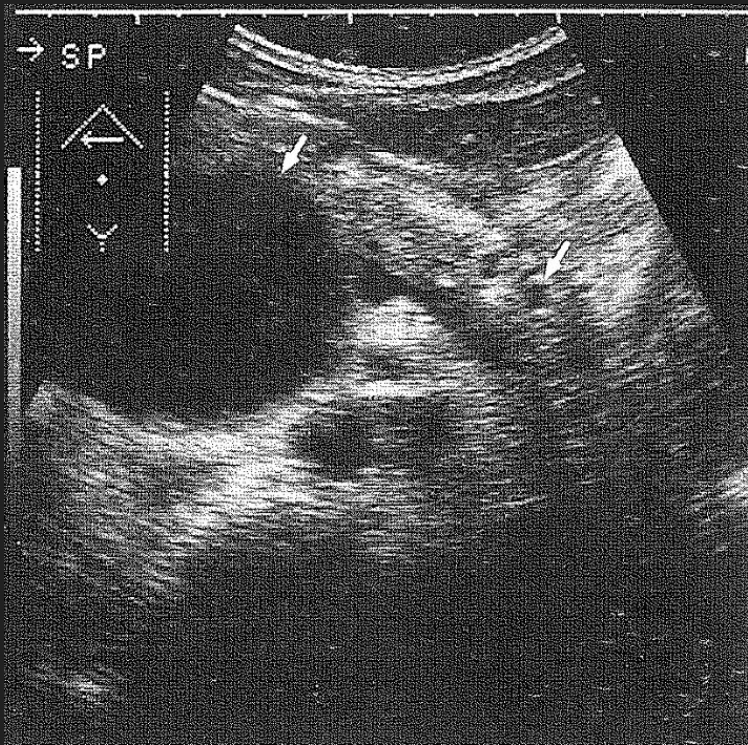
CBD stone



發燒 - 都叫 IAI 就對了啦

- Abscess
 - Liver, Spleen, Kidney, Enteric
- Biliary tract infection
 - Cholecystitis, CBD stone cholangitis
- **Infected fluid accumulation**
 - Peritonitis, Pseudocyst, Infected cyst

Pseudocyst



第二篇章：值班除了切 CT 還能怎麼辦

Fever

Abdominal distention

Abdominal tenderness

GI bleeding

腹脹 - 是水是氣是大便？

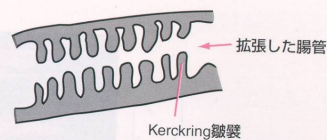
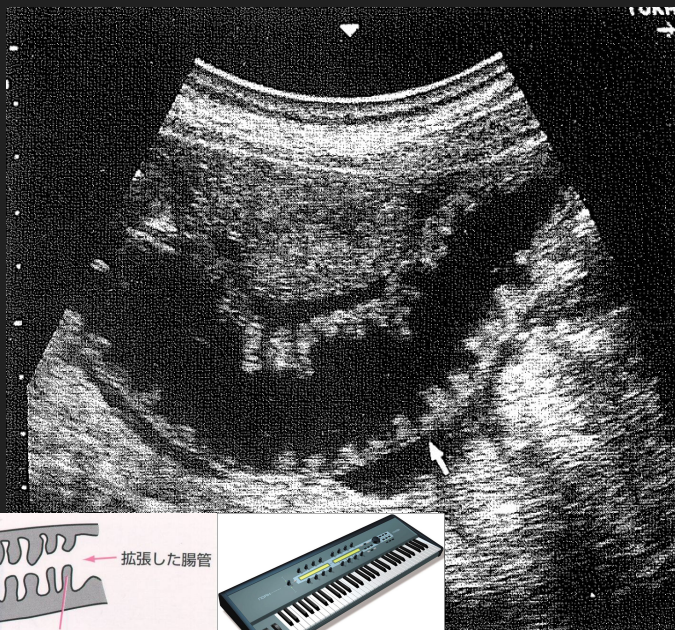
- **Gas** -Tympanic
 - Intraluminal
 - Extraluminal
- **Fluid** -Dullness
 - Ascites (transudate, exudate, bloody)
- **Mass** -Hardness
 - Tumor
 - Organs

Ileus

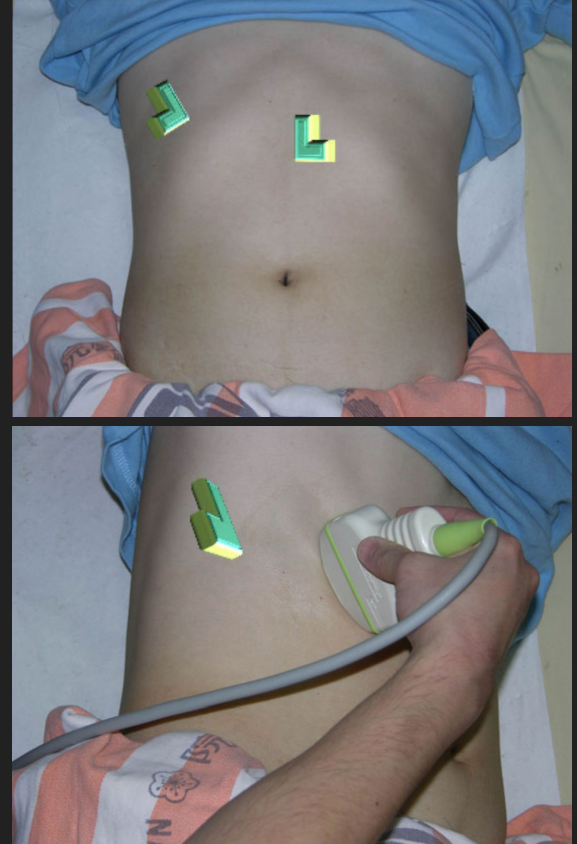
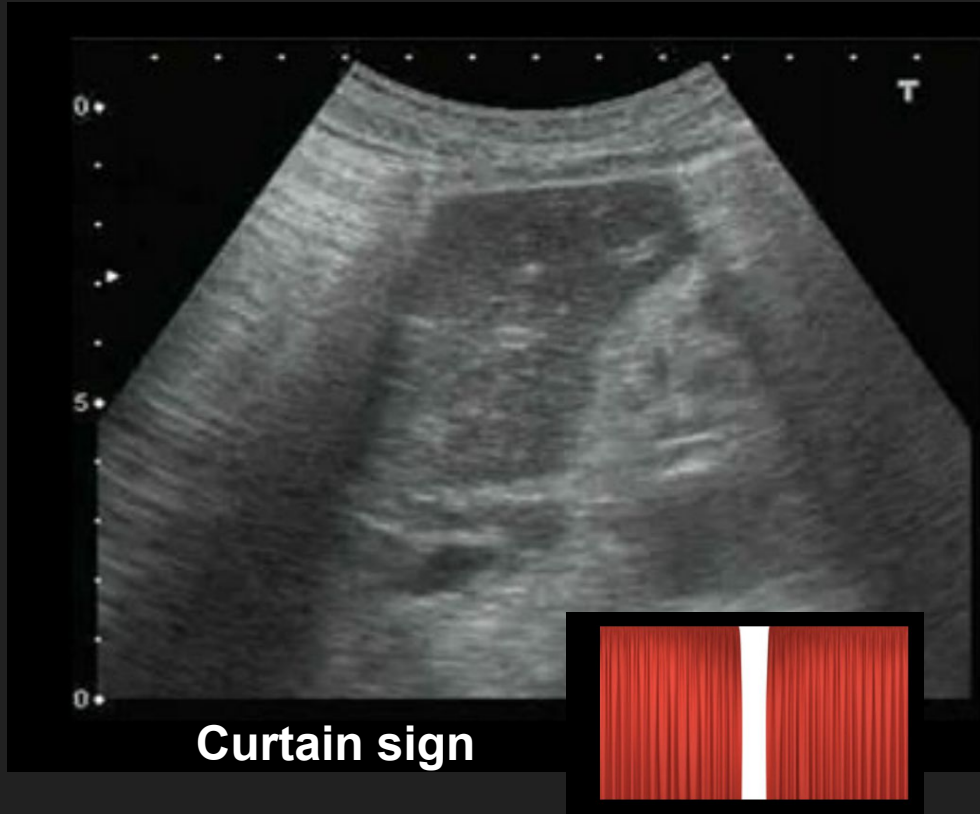
small or large bowel?

paralytic or mechanical?

- 1 可看到腸液來回動的現象
- 2 小腸絞塞：Kerckring 皺壁， Keyboard sign
- 3 大腸絞塞：Haustra

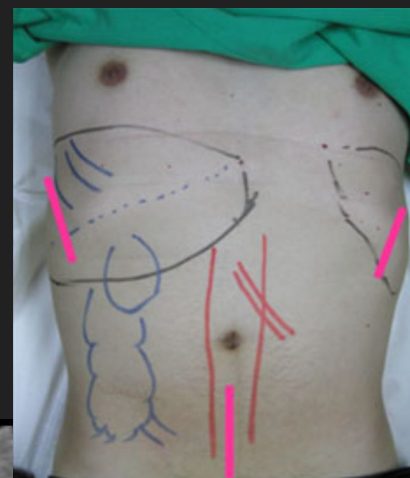


Hollow organ perforation



Ascites

exudate, fibrinous



Ascites

hemoperitoneum

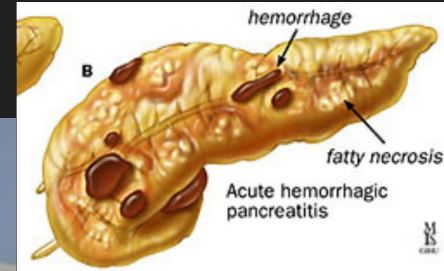


- HCC tumor rupture
- Post-procedural (transfemoral intervention)
- Hemorrhagic pancreatitis
- Aneurysm
- Splenic rupture



Ascites

hemoperitoneum



Cullen's sign (periumbilical ecchymosis)
Grey Turner's sign (flank ecchymosis)

第二篇章：值班除了切 CT 還能怎麼辦

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GI bleeding

腹痛 - 還是乖乖切 CT 吧

- **Inflammation**

- Appendicitis

- **Vascular**

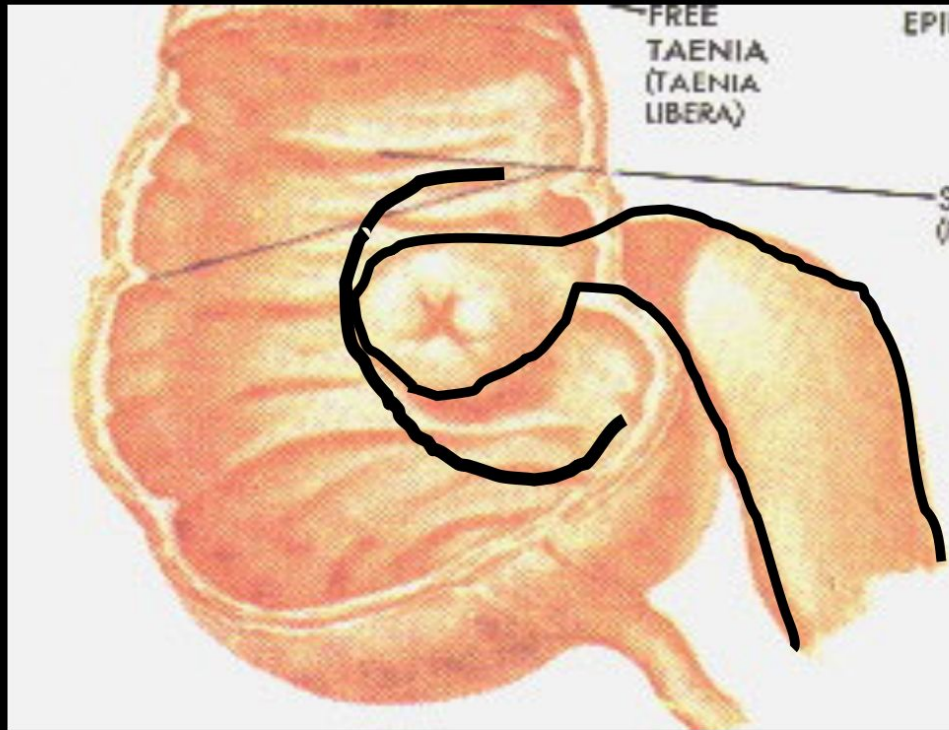
- Aortic dissection
- Ischemic bowel

- **Perforation**

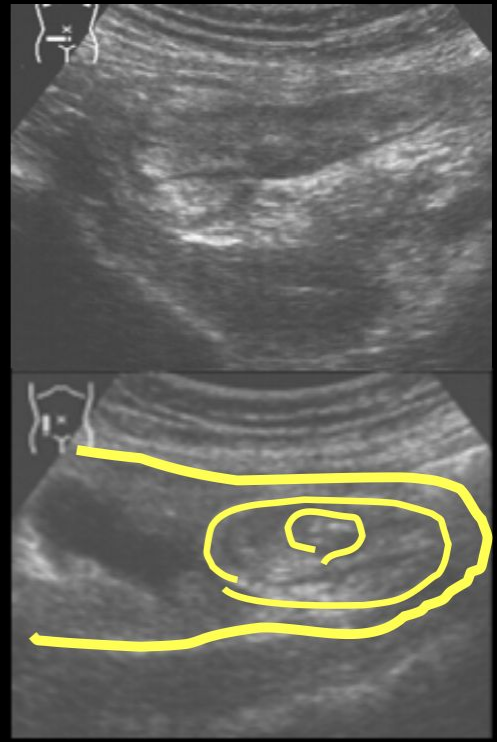
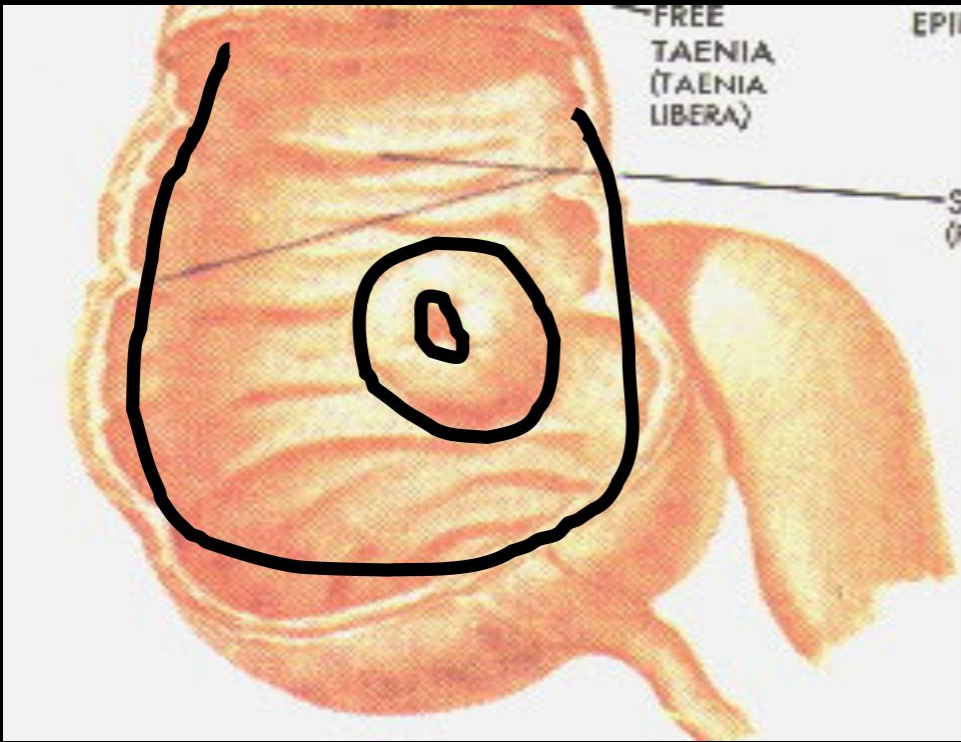
- Hollow organ perforation

- **Obstruction**

- Mechanical ileus



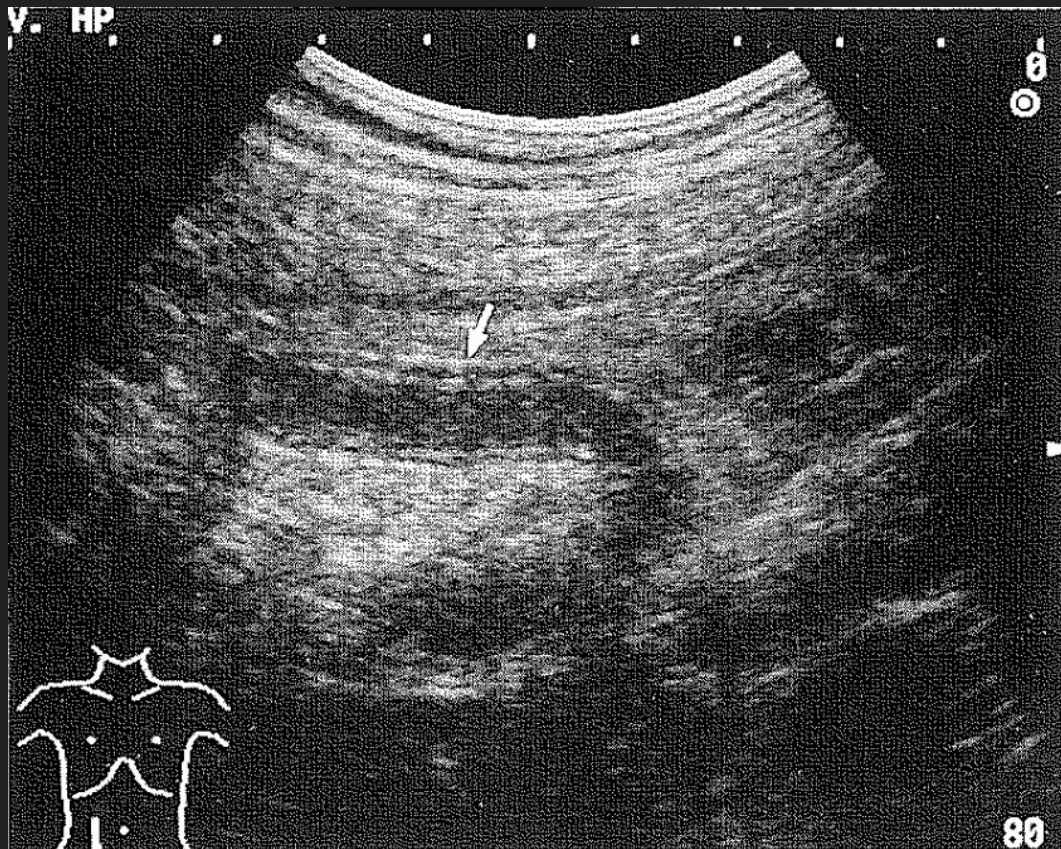
**Sonographic anatomy
of ileal-cecal area**



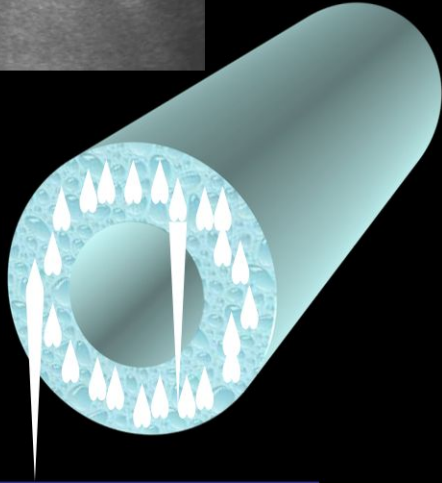
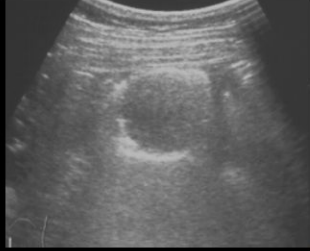
**Sonographic anatomy
of ileal-cecal area**

Acute appendicitis

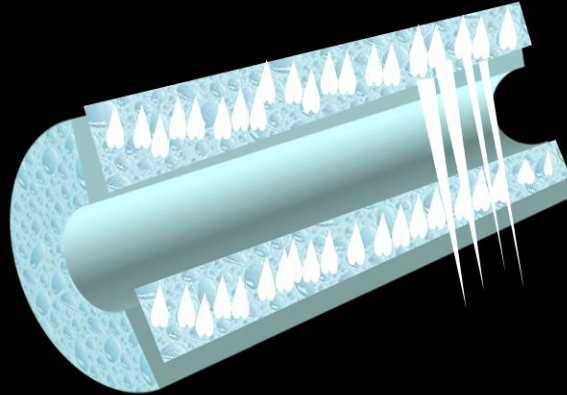
- 1 右下腹部有囊狀～低回音腫瘤影像
- 2 周圍貯留液（腹水，膿瘍）
- 3 腸結石，音陰



Ischemic bowel pneumonitis intestinalis



Circle sign



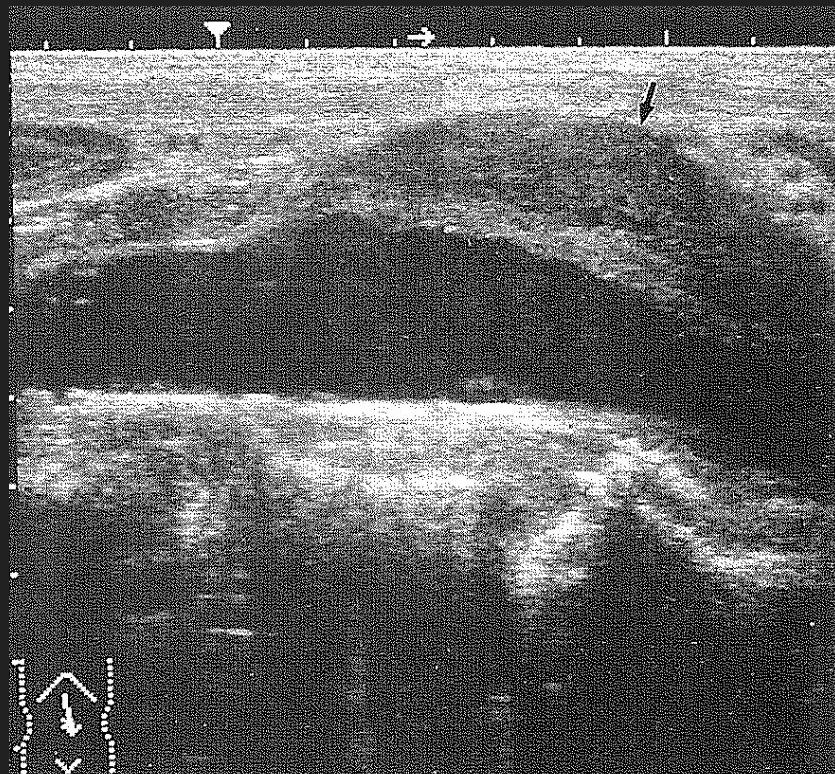
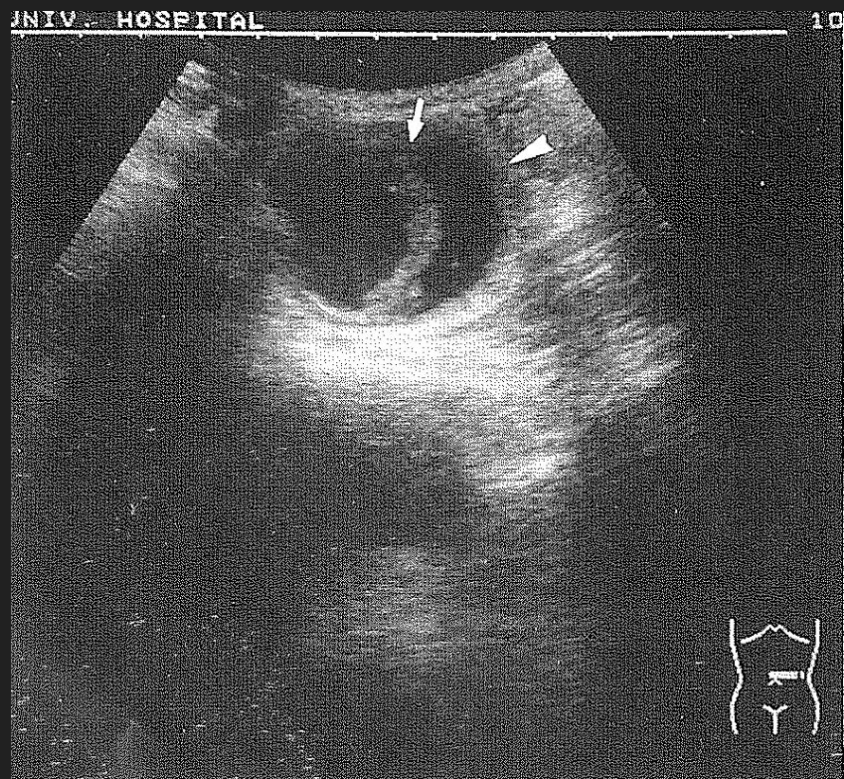
Aurora sign

Ischemic bowel pneumonitis intestinalis



**Aeroportalia sign
(Portal vein gas)**

Aortic Dissection



第二篇章：值班除了切 CT 還能怎麼辦

Fever

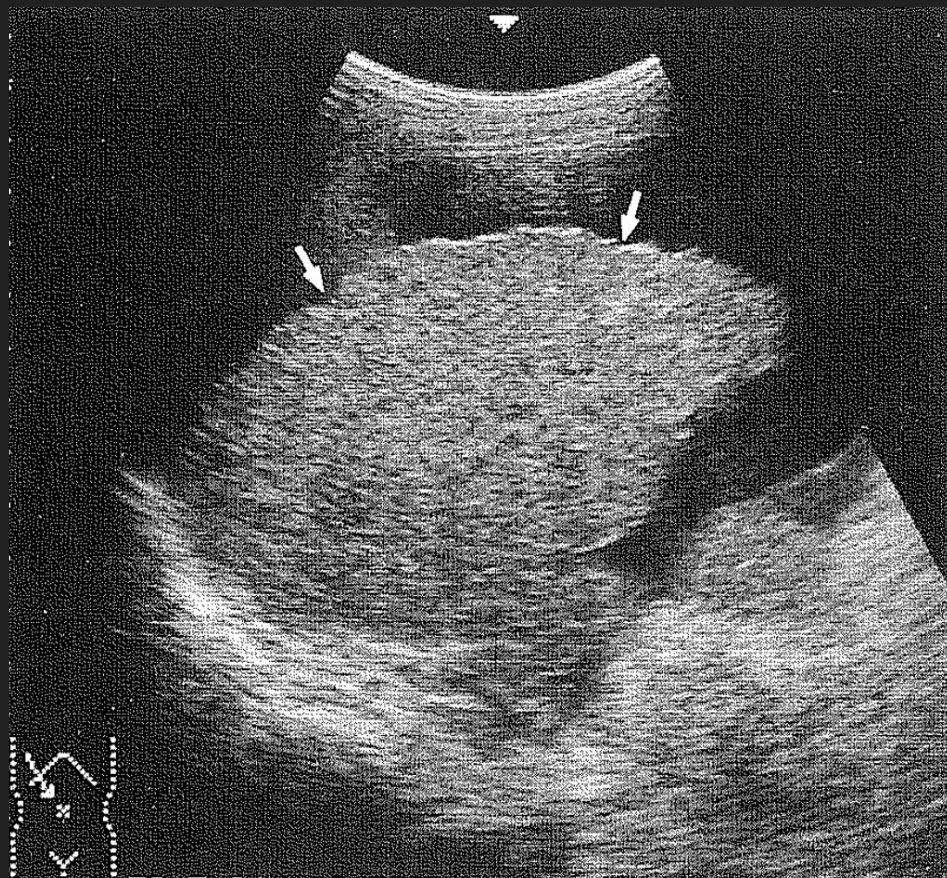
Abdominal distention

Abdominal tenderness

GI bleeding

Liver cirrhosis

- 1 肝表面凹凸不平 3mm
- 2 肝緣鈍化
- 3 肝實質回音粗雜化 (coarse)
- 4 肝內纖維化影像(intrahepatic fibrosis)
- 5 肝內血管彎曲、狹窄、不清晰
- 6 肝右葉萎縮
- 7 肝左葉(尾狀葉, caudate lobe)肥大
- 8 腹水
- 9 膽囊壁肥厚
- 10 肝門靜脈擴張, 蛇行 >14mm
- 11 形成側支循環
- 12 脾臟腫大



診斷肝硬化最有用的三項指標

	Nodular Surface	Coarse Echotexture	Irregularly Narrowed Right Hepatic Vein
Sensitivity	75.5%	83.7%	75.5%
Specificity	99.3%	98.6%	100%
Accuracy	95.9%	96.5%	96.5%
(+) predictive value	94.9%	91.1%	100%
(-) predictive value	96.1%	97.3%	96.1%



Nodular surface

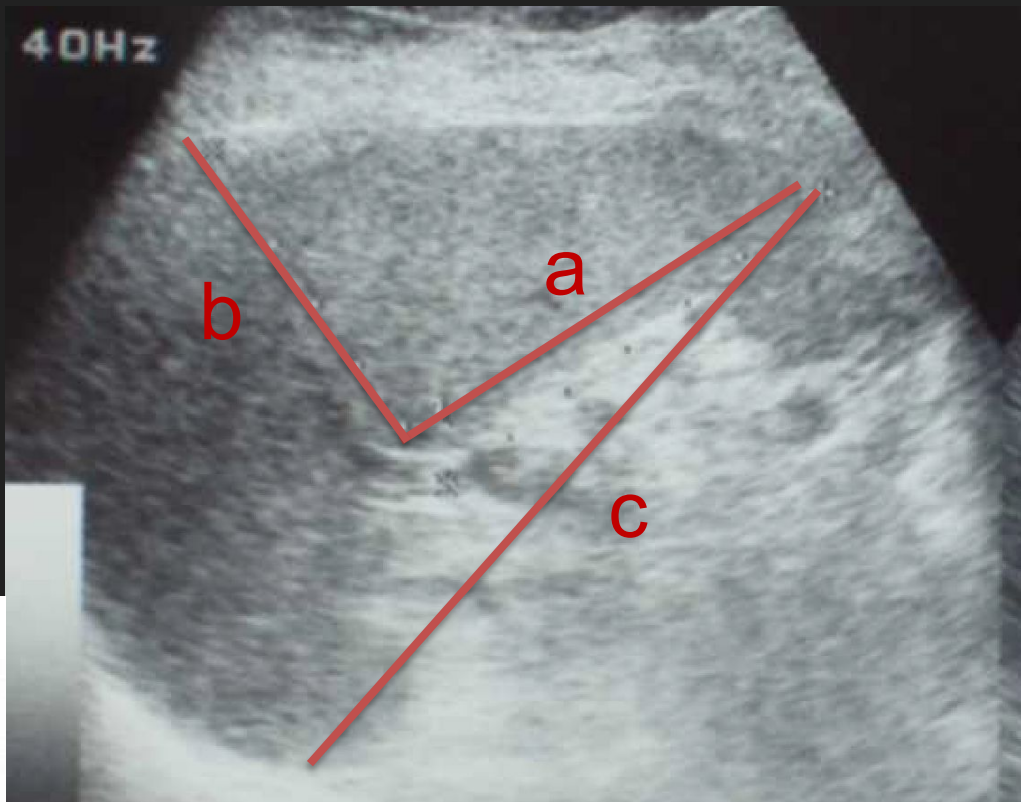


Coarse texture

Blurred tortuous vessel



Splenomegaly

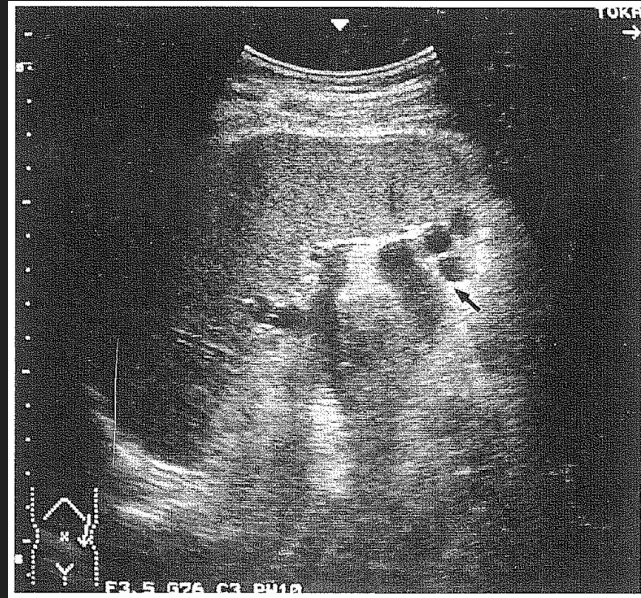
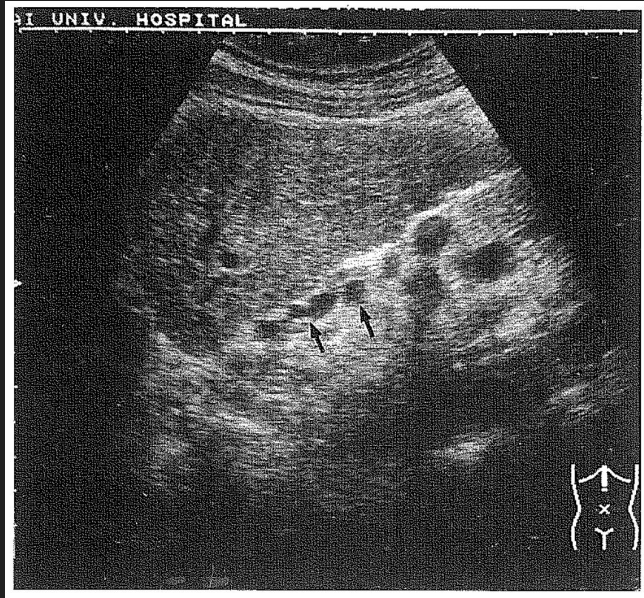


由 **spleen index** 判定脾臟是否腫大

$(a) \times (b) > 20\text{cm}^2$ 千葉大學第一內科的模式

$(c) \times (b) > 30\text{cm}^2$ 古賀模式

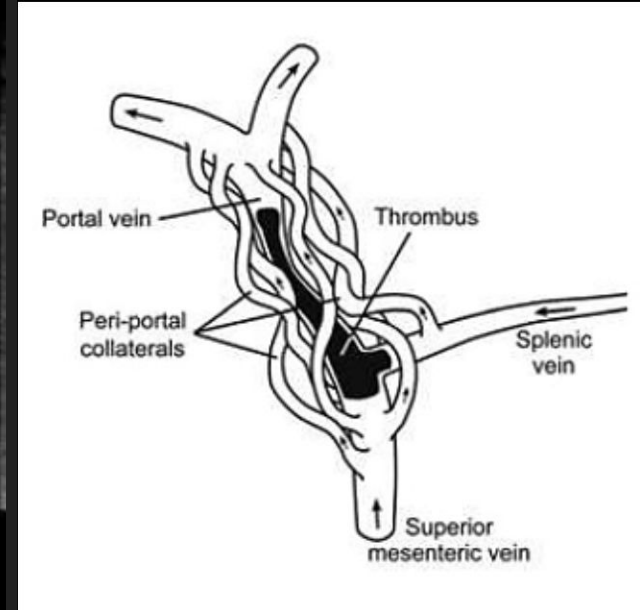
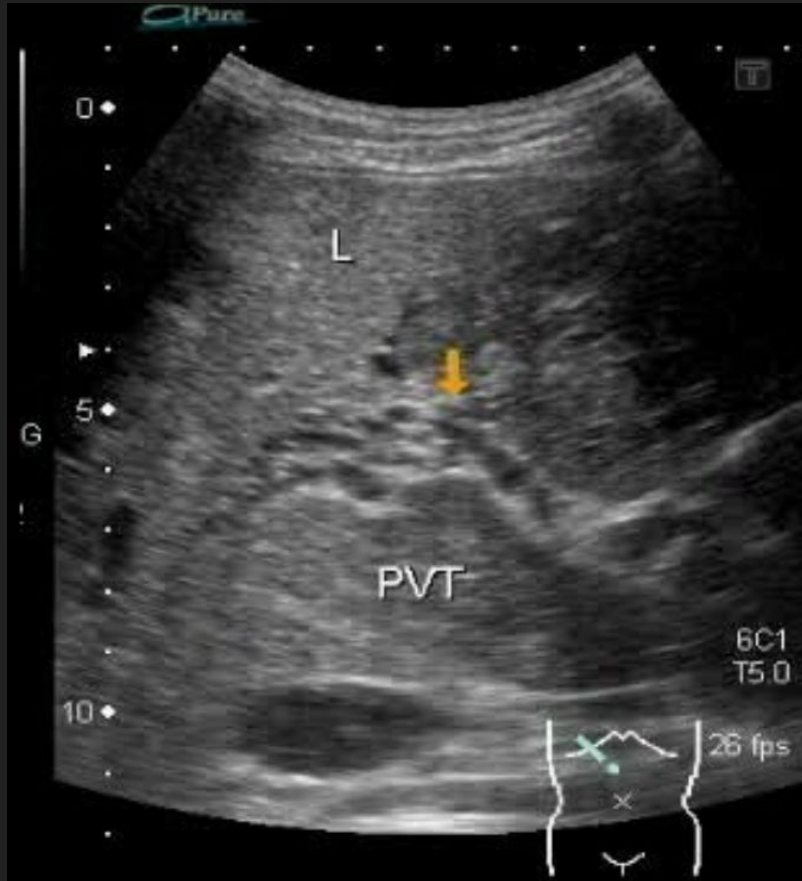
Collaterals



Cavernous transformation of portal vein



Diamond sign



Gastric ultrasound for risk of aspiration



Fever

Abscess, Biliary tract infection, Infected fluid

Abdominal distention

Ileus, Perforation, Ascites, Hemoperitoneum

Abdominal tenderness

Inflammation, Perforation, Vascular, Obstruction

GI bleeding

Liver cirrhosis, Gastric varices, Aspiration risk

